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Review

Path Planning for Autonomous Mobile Robots: A Review

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Abstract: Providing mobile robots with autonomous capabilities is advantageous. It allows one to dispense with the intervention of human operators, which may prove beneficial in economic and safety terms. Autonomy requires, in most cases, the use of path planners that enable the robot to deliberate about how to move from its location at one moment to another. Looking for the most appropriate path planning algorithm according to the requirements imposed by users can be challenging, given the overwhelming number of approaches that exist in the literature. Moreover, the past review works analyzed here cover only some of these approaches, missing important ones. For this reason, our paper aims to serve as a starting point for a clear and comprehensive overview of the research to date. It introduces a global classification of path planning algorithms, with a focus on those approaches used along with autonomous ground vehicles, but is also extendable to other robots moving on surfaces, such as autonomous boats. Moreover, the models used to represent the environment, together with the robot mobility and dynamics, are also addressed from the perspective of path planning. Each of the path planning categories presented in the classification is disclosed and analyzed, and a discussion about their applicability is added at the end.

Keywords: guidance; autonomy; vehicle; survey; trajectory; route; graph search; sampling; wheeled

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1. Introduction

Autonomous navigation is a valuable asset for mobile robots. It helps to mitigate their dependency on human intervention. However, it also entails many tasks or problems to solve, e.g., path planning. This task lies in finding the best course of action to make a robot reach the desired state from its current one. For example, both states could be, respectively, the goal and the initial position. This course of action comes in the form of a path, also named a route in many other works. The path serves to guide the robot to the desired state in question. However, there may be numerous possible paths, given the free space in which the robot can move. Path planning algorithms generally try to obtain the best path or at least an admissible approximation to it. The best path here refers to the optimal one, in the sense that the resulting path comes from minimizing one or more objective optimization functions. For instance, this path may be the one entailing the least amount of time. This is critical in missions such as those of the search-and-rescue field [1]: victims of a disaster may ask for help in life-or-death situations. Another optimization function to consider could be the energy of the robot. In the case of planetary exploration, this is critical since rovers have limited energetic resources available [2]. At the same time, the path generated by the planner must follow any imposed restrictions. These may come from the limitations in the adaptability of the robot to certain terrains. The locomotion of the robot and the characteristics of the existing terrain limit the kind of manoeuvres that can be performed. This consequently reduces the number of feasible paths that the path planner can generate.

In the literature there are a vast number of path planning approaches and this number has continued to increase over the years. For this reason, selecting the most appropriate approach given certain requirements (for instance, the aforementioned locomotion restrictions) can be a challenging task. Moreover, as discussed below, the latest reviews

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Review

自主移动机器人的路径规划：回顾

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摘要：为移动机器人提供自主能力是有利的。它允许人们免除人工操作员的干预，这在经济和安全方面可能是有益的。在大多数情况下，自主性需要使用路径规划器，使机器人能够考虑如何从某一时刻的位置移动到另一时刻。鉴于文献中存在大量方法，根据用户的要求寻找最合适的路径规划算法可能具有挑战性。此外，这里分析的过去的评论工作仅涵盖了其中的一些方法，遗漏了重要的方法。因此，我们的论文旨在作为一个起点，对迄今为止的研究进行清晰、全面的概述。它引入了路径规划算法的全局分类，重点关注与自主地面车辆一起使用的方法，但也可扩展到在水面上移动的其他机器人，例如自主船。此外，还从路径规划的角度解决了用于表示环境的模型以及机器人的移动性和动力学。分类中提出的每个路径规划类别都被公开和分析，并在最后添加了关于它们的适用性的讨论。

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关键词：指导；自治;车辆;民意调查;弹道;路线;图搜索；采样;轮式的

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一、简介

自主导航对于移动机器人来说是一项宝贵的资产。它有助于减轻他们对人类干预的依赖。然而，它也需要解决许多任务或问题，例如路径规划。这项任务在于找到使机器人从当前状态达到所需状态的最佳行动方案。例如，两个状态可以分别是目标位置和初始位置。这个行动过程以路径的形式出现，在许多其他作品中也被称为路线。该路径用于引导机器人到达所讨论的所需状态。然而，考虑到机器人可以移动的自由空间，可能有许多可能的路径。路径规划算法通常尝试获得最佳路径或至少是可接受的近似路径。这里的最佳路径是指最优路径，即所得路径来自最小化一个或多个目标优化函数。例如，这条路径可能是花费最少时间的路径。这对于搜救领域的任务至关重要 [1]：灾难受害者可能会在生死攸关的情况下寻求帮助。另一个需要考虑的优化函数是机器人的能量。在行星探索的情况下，这一点至关重要，因为漫游车可用的能量资源有限[2]。同时，规划器生成的路径必须遵循任何强加的限制。这些可能来自于机器人对某些地形的适应性的限制。机器人的运动和现有地形的特征限制了可以执行的机动类型。因此，这减少了路径规划器可以生成的可行路径的数量。

文献中有大量的路径规划方法，而且这个数字多年来一直在持续增加。因此，在给定某些要求（例如上述的运动限制）的情况下选择最合适的方法可能是一项具有挑战性的任务。此外，如下所述，最新评论

and surveys on path planning do not offer a comprehensive overview of the majority of existing path planning solutions. This is the main motivation for writing this review paper: it describes in detail different path planning categories and, for each of them, introduces relevant representative references found in the literature, focusing on those algorithms aimed at robots that move on top of surfaces (ground, water, etc.). This paper is organized as follows. Section 2 presents the method proposed in this paper to classify the existing path planning algorithms. It also makes clear, following this method, the fact that previous works have important omissions. Moreover, this section also provides an analysis of the methods used to address the environment and locomotion information. The next sections each deal with one of the categories of this classification: *Reactive Computing* (Section 3), *Soft Computing* (Section 4), *C-Space Search* (Section 5) and *Optimal Control* (Section 6). Finally, Section 7 summarizes the contents of this paper and presents a discussion about the path planning algorithms contained in the aforementioned categories.

2. Path Planning Algorithms

Figure 1 depicts the four mentioned categories of path planning, splitting each of them into two subcategories. This classification rests on the principles and fundamental mechanisms used to construct and return a path. A more detailed insight into these categories and why this classification is arranged in this way is provided in the next subsection. A second subsection presents the different approaches taken in modelling the environment and the robot–terrain interaction. We considered it necessary to add this as for many algorithms, especially those in the *C-Space Search* category, it is necessary to construct these models beforehand.

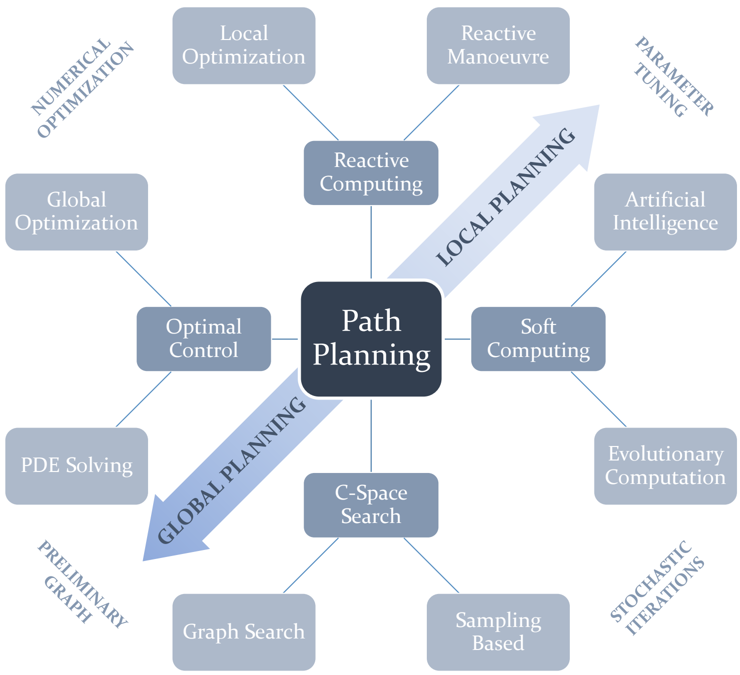


Figure 1. Schematic showing the proposed classification of existing path planning approaches. There are four main categories, each of them containing two subcategories. Two adjacent subcategories from different categories have features in common. The schematic also indicates how some subcategories are more inclined towards either *Global Planning* or *Local Planning*.

2.1. General Classification

The proposed classification considers how path planning algorithms function. In many past reviews, two kinds of distinction were made: according to whether the environment is dynamic or not; *Online* and *Offline* path planners. respectively [3]; and the size of the environment, whether *Local* or *Global*. Usually, *Online* is associated with *Local* and *Offline* with *Global*. The main issue with this is that there are algorithms that can be considered in

关于路径规划的调查并没有提供对大多数现有路径规划解决方案的全面概述。这是撰写这篇综述论文的主要动机：它详细描述了不同的路径规划类别，并针对每个类别介绍了文献中的相关代表性参考文献，重点关注针对在表面（地面、水面等）上移动的机器人的算法。本文的结构如下。第 2 部分介绍了本文提出的对现有路径规划算法进行分类的方法。按照这种方法，也清楚地表明以前的作品有重要的遗漏。此外，本节还分析了用于处理环境和运动信息的方法。接下来的各节分别涉及此分类的一个类别：*Reactive Computing* (第 3 节)、*Soft Computing* (第 4 节)、*C-Space Search* (第 5 节) 和 *Optimal Control* (第 6 节)。最后，第七节总结了本文的内容，并对上述类别中包含的路径规划算法进行了讨论。

2. 路径规划算法

图 1 描述了路径规划的四个类别，并将每个类别分为两个子类别。这种分类基于用于构建和返回路径的原则和基本机制。下一小节将更详细地介绍这些类别以及为什么以这种方式安排这种分类。第二小节介绍了对环境和机器人与地形交互进行建模时所采用的不同方法。我们认为有必要添加这一点，因为对于许多算法，尤其是*C-Space Search*类别中的算法，有必要预先构建这些模型。

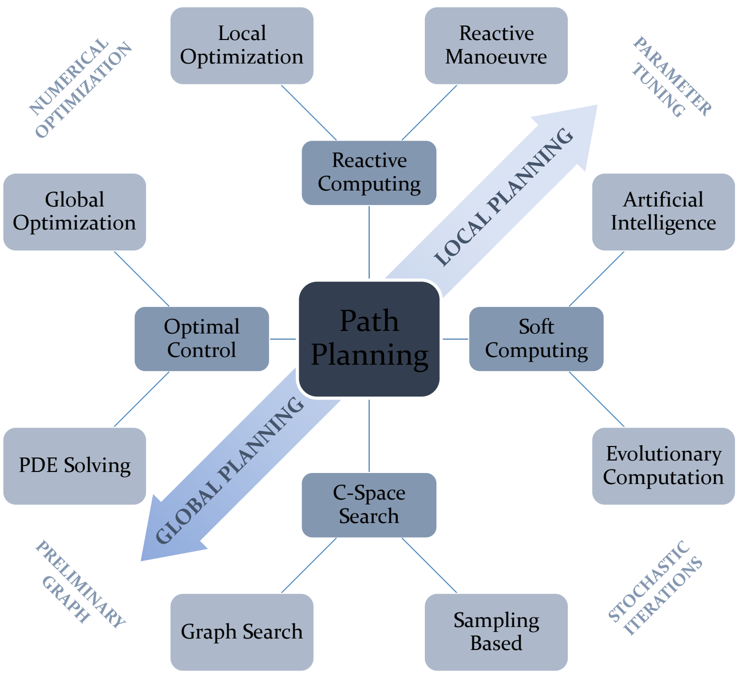


图 1. 示意图显示了现有路径规划方法的拟议分类。有四个主要类别，每个类别又包含两个子类别。不同类别的两个相邻子类别具有共同特征。该示意图还表明某些子类别如何更倾向于 *Global Planning* 或 *Local Planning*。

2.1. General Classification

所提出的分类考虑了路径规划算法的运作方式。在过去的许多评论中，有两种区分：根据环境是否动态；以及根据环境是否动态。 *Online* 和 *Offline* 路径规划器。分别[3]；以及环境的大小，无论是 *Local* 还是 *Global*。通常，*Online* 与 *Local* 关联，*Offline* 与 *Global* 关联。主要问题是有一些算法可以考虑

both categories. An algorithm with no replanning capabilities could be used online due to its high computational speed. The contrary could also happen. For instance, a *Reactive Computing* algorithm called the Dynamic Window Approach (DWA), is usually used for local planning [4], but can also be used for global planning [5]. Vagale et al. [6] presents an interesting division between algorithms that require a preliminary map representation (*Classic*) [7] and those which do not (*Advanced*). *Classic* includes *Graph Search* methods, whereas *Advanced* addresses *Soft Computing* and *Sampling-Based* algorithms. Souissi et al. [7] proposes several clear and reasonable path planning classifications: according to the robot model (holonomic, non-holonomic, kinodynamic); according to the map model requirement (needed or not needed beforehand); according to the replanning capability (offline or online); and according to whether the algorithm always calculates the same solution or not, according to preliminary configuration parameters (deterministic or probabilistic).

The main purpose of the classification proposed in this paper, depicted in Figure 1, is two-fold. First, this classification aims to encompass a larger variety of algorithms than those that are tackled in past reviews. Many past reviews propose or claim to present a general overview on path planning, but as can be seen in Table 1 the majority of them suffer from important omissions. In this table, *Yes* means there is significant discussion about the algorithms in question. *Only Mentions* means the publication acknowledges the existence of at least one or more algorithms in that class. If there are one or two algorithms between parentheses, this means those are only mentioned/referred to briefly. Second, the nomenclature to refer to path planning categories is not clear in some cases. For instance, some other reviews make a distinction between *Classical* and *Heuristic* approaches [8,9]. Patle et al. [10] refer to the latter as *Reactive*. However, the term *Classical* can be quite an ambiguous term as the majority of planning algorithms used are based on methods with one or more decades of life. This class is also used to encompass many algorithms with completely different ways of functioning. The term *Heuristic* has not only been used to refer to *Evolutionary* and *Artificial Intelligence* algorithms [8], but it has also been also used to refer to *Graph Search*-based planners [4]. With this in mind, we propose the use of a general classification using four classes (see Figure 1): *Reactive Computing*, *Soft Computing*, *C-Space Search*, and *Optimal Control*. Moreover, Figure 1 shows how in general terms each of these categories tend to be used mostly for either *Local* or *Global planning*. Moreover, each of the subcategories can also have something in common in their functioning with subcategories from other categories, such as the use of numerical methods, the existence of parameters to tune the algorithm beforehand, the requirement of modelling the map with a graph or the use of stochastic iterative processes.

Table 1. Main surveys and reviews of global path planning algorithms found in the literature.

	Reactive Computing		Soft Computing		C-Space Search		Optimal Control	
Publication	**Reactive Manoeuvre**	**Local Optimization**	**Evolutionary Computation**	**Artificial Intelligence**	**Sampling Based**	**Graph Search**	**PDE Solving**	**Global Optimization**
Raja and Pugazhenthi [3] (2012)	Yes	No	Yes	No	No	No	No	No
Nash and Koenig [11] (2013)	No	No	No	No	Yes (RRT and PRM)	Yes (Any-angle and A*)	No	No
Souissi et al. [7] (2013)	Yes (APF)	No	Only mentions	No	Yes	Yes	No	No
Elbanhawi and Simic [12] (2014)	Only mentions	No	Only mentions	Only mentions	Yes	Only mentions	No	No
González et al. [13] (2015)	No	Yes	No	No	Yes	Yes	No	Yes (NLP)
Mac et al. [8] (2016)	Yes	No	Yes	Yes	Yes	No	No	No
Noreen et al. [14] (2016)	No	No	Only mentions	No	Yes	Only mentions	No	No
Injarapu and Gawre [15] (2017)	No	No	Yes (GA)	Yes	No	No	No	No

两个类别。没有重新规划能力的算法由于其高计算速度可以在线使用。相反的情况也可能发生。例如，称为动态窗口方法 (DWA) 的 *Reactive Computing* 算法通常用于局部规划 [4]，但也可用于全局规划 [5]。瓦加莱等人。 [6] 提出了需要初步地图表示 (*Classic*) [7] 和不需要初步地图表示 (*Advanced*) 的算法之间的有趣划分。 *Classic* 包括 *Graph Search* 方法，而 *Advanced* 则解决 *Soft Computing* 和 *Sampling-Based* 算法。苏伊西等人。 [7]提出了几种清晰合理的路径规划分类：根据机器人模型（完整、非完整、运动动力学）；根据地图模型要求（事先需要或不需要）；根据重新规划能力（离线或在线）；并根据算法是否总是计算出相同的解，根据初步配置参数（确定性或概率性）。

本文提出的分类的主要目的如图 1 所示，有两个目的。首先，这种分类旨在涵盖比过去评论中处理的算法更多种类的算法。过去的许多评论提出或声称对路径规划进行了总体概述，但从表 1 中可以看出，大多数评论都存在重要的遗漏。在此表中，*Yes* 表示对相关算法进行了重要讨论。 *Only Mentions* 表示该出版物承认该类中至少存在一种或多种算法。如果括号之间有一种或两种算法，则意味着这些算法仅被简要提及/提及。其次，在某些情况下，路径规划类别的术语并不明确。例如，其他一些评论对 *Classical* 和 *Heuristic* 方法进行了区分 [8,9]。帕特尔等人。 [10]将后者称为*Reactive*。然而，术语 *Classical* 可能是一个相当模糊的术语，因为大多数使用的规划算法都基于具有十年或几十年寿命的方法。此类还用于包含许多具有完全不同功能方式的算法。术语 *Heuristic* 不仅用于指代 *Evolutionary* 和 *Artificial Intelligence* 算法 [8]，而且还用于指代基于 *Graph Search* 的规划器 [4]。考虑到这一点，我们建议使用四个类别的通用分类（参见图 1）： *Reactive Computing*、*Soft Computing*、*C-Space Search* 和 *Optimal Control*。此外，图 1 显示了一般而言，这些类别中的每一个往往主要用于 *Local* 或 *Global planning*.。此外，每个子类别在功能上也可以与其他类别的子类别有一些共同点，例如数值方法的使用、预先调整算法的参数的存在、使用图形对地图进行建模的要求或随机迭代过程的使用。

表 1. 文献中对全局路径规划算法的主要调查和评论。

	Reactive Computing		**Soft Computing**		**C-Space Search**		**Optimal Control**	
Publication	**Reactive Manoeuvre**	**Local Optimization**	**Evolutionary Computation**	**Artificial Intelligence**	**Sampling Based**	**Graph Search**	**PDE Solving**	**Global Optimization**
Raja and Pugazhenthi [3] (2012)	Yes	No	Yes	No	No	No	No	No
Nash and Koenig [11] (2013)	No	No	No	No	Yes (RRT and PRM)	Yes (Any-angle and A*)	No	No
Souissi et al. [7] (2013)	Yes (APF)	No	Only mentions	No	Yes	Yes	No	No
Elbanhawi and Simic [12] (2014)	Only mentions	No	Only mentions	Only mentions	Yes	Only mentions	No	No
González et al. [13] (2015)	No	Yes	No	No	Yes	Yes	No	Yes (NLP)
Mac et al. [8] (2016)	Yes	No	Yes	Yes	Yes	No	No	No
Noreen et al. [14] (2016)	No	No	Only mentions	No	Yes	Only mentions	No	No
Injarapu and Gawre [15] (2017)	No	No	Yes (GA)	Yes	No	No	No	No

Table 1. *Cont.*

	Reactive Computing		Soft Computing		C-Space Search		Optimal Control	
Publication	**Reactive Manoeuvre**	**Local Optimization**	**Evolutionary Computation**	**Artificial Intelligence**	**Sampling Based**	**Graph Search**	**PDE Solving**	**Global Optimization**
Ravankar et al. [16] (2018)	No	Yes	No	No	No	No	No	Yes (NLP)
Zhang et al. [4] (2018)	Yes	No	Yes	Yes (ANN)	No	Yes	No	No
Zafar and Mohanta [9] (2018)	Yes (APF)	No	Yes	Yes (ANN)	Yes	Yes	No	No
Costa and Silva [17] (2019)	No	No	Yes (GA)	No	Yes (RRT)	Yes (A*)	No	No
Patle et al. [10] (2019)	Yes (APF)	No	Yes	Yes	Only mentions (PRM)	No	No	No
Gómez et al. [18] (2019)	No	No	No	No	No	No	Yes (eikonal solvers)	No
Campbell et al. [19] (2020)	Yes	No	Yes	Yes	No	No	No	No
Zhang et al. [20] (2020)	Yes	No	Yes	No	Yes	Yes (A* Variants)	Yes (Fast Marching)	Yes (DP)
Sun et al. [21] (2021)	Only mentions (APF, DWA)	No	Only mentions	Yes	Only mentions (RRT and PRM)	Only mentions	No	No
Vagale et al. [6] (2021)	Yes	No	Yes	Yes (DRL)	Yes (RRT and PRM)	No	No	No

2.2. Path Planning Workspace Modeling

A path planner needs to be fed with information describing the environment. This information can describe, for example, the presence of obstacles or the features of the surface that are relevant to the planning. In addition, the criteria used to calculate the path has to do with the way the robot interacts with this environment. For instance, merely to minimize path length and perhaps acquire the information as to what areas can be traversed or not is enough, whereas to minimize energy the terramechanics and the way the robot steers should be taken into account.

2.2.1. Environment Modeling

Surface mobile robots drive from one position to another within a certain region in space. Therefore, it is necessary to consider how the locomotion model will interact with this surface and how the path planner will take care of it. For instance, some algorithms require the construction of a graph that somehow represents the environment in which the robot is moving. This is mostly the case for *Graph Search* algorithms, part of the *C-Space Search* category. *Evolutionary* algorithms such as Ant Colony Optimizers (ACO) can also make use of a graph. This asset can represent how the terrain features that affect the robot navigation are spatially arranged in the scenario. In particular, the graph in question is assumed here to be built upon metric maps, acknowledging the existence of other kinds of maps outside of the scope of this paper, such as topological and semantic maps [22]. According to Souissi et al. [7] there are multiple ways to build a graph, as shown in Figure 2. The work of Nash and Koenig [11] has also shed some light onto this classification. It distinguishes between *Cell Decomposition* and *Roadmaps*. The first of these consists of tessellating the surface into cells. These cells can be arranged using regular [3,4,7,11] or irregular grids [7,11]. Figures 2a–c show how regular grids can be built using one out of three types of polygons: squares, triangles and hexagons. Its main advantage is the simple indexation of each node, which is translated into quick access to any of them and an optimized way to store them in memory [23]. Irregular grids, such as the one depicted in Figure 2d, allow the better adaptation the grid to terrain features with

表 1. *Cont.*

	Reactive Computing		Soft Computing		C-Space Search		Optimal Control	
Publication	**Reactive Manoeuvre**	**Local Optimization**	**Evolutionary Computation**	**Artificial Intelligence**	**Sampling Based**	**Graph Search**	**PDE Solving**	**Global Optimization**
Ravankar et al. [16] (2018)	No	Yes	No	No	No	No	No	Yes (NLP)
Zhang et al. [4] (2018)	Yes	No	Yes	Yes (ANN)	No	Yes	No	No
Zafar and Mohanta [9] (2018)	Yes (APF)	No	Yes	Yes (ANN)	Yes	Yes	No	No
Costa and Silva [17] (2019)	No	No	Yes (GA)	No	Yes (RRT)	Yes (A*)	No	No
Patle et al. [10] (2019)	Yes (APF)	No	Yes	Yes	Only mentions (PRM)	No	No	No
Gómez et al. [18] (2019)	No	No	No	No	No	No	Yes (eikonal solvers)	No
Campbell et al. [19] (2020)	Yes	No	Yes	Yes	No	No	No	No
Zhang et al. [20] (2020)	Yes	No	Yes	No	Yes	Yes (A* Variants)	Yes (Fast Marching)	Yes (DP)
Sun et al. [21] (2021)	Only mentions (APF, DWA)	No	Only mentions	Yes	Only mentions (RRT and PRM)	Only mentions	No	No
Vagale et al. [6] (2021)	Yes	No	Yes	Yes (DRL)	Yes (RRT and PRM)	No	No	No

2.2. *Path Planning Workspace Modeling*

路径规划器需要提供描述环境的信息。例如，该信息可以描述与规划相关的障碍物的存在或表面特征。此外，用于计算路径的标准与机器人与该环境交互的方式有关。例如，仅仅为了最小化路径长度并可能获取关于可以遍历或不遍历哪些区域的信息就足够了，而为了最小化能量，应该考虑地形力学和机器人的转向方式。

2.2.1.环境建模

地面移动机器人在空间某一区域内从一个位置行驶到另一个位置。因此，有必要考虑运动模型如何与该表面交互以及路径规划器将如何处理它。例如，某些算法需要构建一个图表来以某种方式表示机器人移动的环境。这主要是 *Graph Search* 算法的情况，它是 *C-Space Search* 类别的一部分。 *Evolutionary* 算法（例如蚁群优化器 (ACO)）也可以使用图。该资源可以表示影响机器人导航的地形特征在场景中的空间排列方式。特别是，这里假设所讨论的图是建立在度量图的基础上的，承认存在本文范围之外的其他类型的图，例如拓扑和语义图[22]。根据 Souissi 等人的说法。 [7] 有多种方法可以构建图，如图 2 所示。Nash 和 Koenig [11] 的工作也为这种分类提供了一些启示。它区分 *Cell Decomposition* 和 *Roadmaps*。第一个是将表面细分为单元。这些单元格可以使用规则[3,4,7,11]或不规则网格[7,11]进行排列。图 2a-c 显示了如何使用三种类型的多边形之一来构建规则网格：正方形、三角形和六边形。它的主要优点是每个节点的简单索引，这可以转化为对任何节点的快速访问以及将它们存储在内存中的优化方式[23]。不规则网格，如图 2d 所示，可以使网格更好地适应地形特征

different values of resolution, at the expense of possibly obtaining worse paths [24]. Other forms of *Cell Decomposition* are the *Navigation meshes* and *Circle-based waypoint graphs*, as explained by Nash and Koenig [11]. The other form of representation of the environment is, as mentioned, accomplished using roadmaps. A roadmap is a graph built upon nodes connected by edges. Each node represents a possible state of the robot, whereas each edge indicates how to reach that state from another. Examples of roadmaps include Voronoi graphs [25] (see Figure 2e), Visibility graphs [26] and State-Lattice graphs (see Figure 2f). The latter consist of making the edges based upon motion primitives, so the resulting path is ensured to be feasible given the robot mobility constraints, especially when using *Graph Search* algorithms, as in the work of Likhachev and Ferguson [27], Bergman et al. [28].

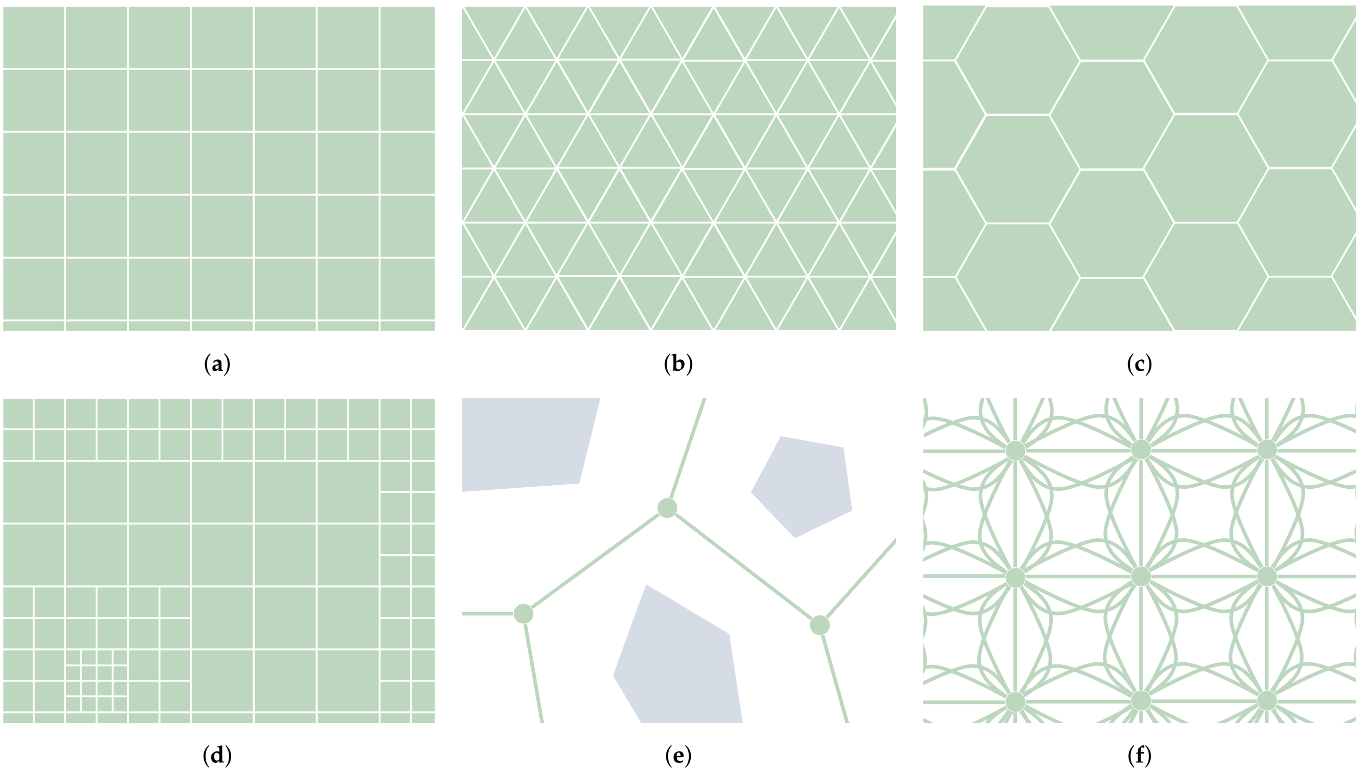


Figure 2. Different types of environment cell decomposition (**a**–**d**) and roadmap graphs (**e**–**f**). (**a**) Tessellations using squares. (**b**) Tessellations using triangles. (**c**) Tessellations using hexagons. (**d**) Irregular grid. (**e**) Voronoi roadmap. (**f**) State lattice graph.

The cells or the nodes from these graphs can store information regarding the surface at their location, in the form of static or dynamic elements [10]. This can be, for instance, elevation information. A Digital Elevation Map (DEM) is a grid of which the nodes have associated with each of them a value of elevation. Elevation maps can be also represented by polygons, but regular grid maps are preferred [29]. Shape-related features, such as the slope or the surface roughness, can be extracted using convolution matrices [30]. The size of the kernel and the DEM resolution will determine what kind of features are extracted. Moreover, this resolution defines the level of detail of the elements contained in the map. As shown in Figure 2d, this resolution can be non-uniform or multiple. The size of the grid can be chosen according to the scale at which the planning is performed: *Local* in the case of covering the immediate surroundings of the robot (more or less the reachable distance of the on-board sensors) and *Global* if the area is bigger than that, usually using information from external sources such as satellites or drones.

With regards to how the cost is defined over the planner workspace, there are different ways. First of all, we understand cost as the metric that the robot accumulates by moving. The objective of the path planner is to minimize this accumulation by producing the optimal path. The cost in question can be uniform, in the sense that the regions that can be accessed

不同的分辨率值，代价是可能获得更差的路径[24]。正如 Nash 和 Koenig [11] 所解释的，*Cell Decomposition* 的其他形式是 *Navigation meshes* 和 *Circle-based waypoint graphs*。如前所述，环境的另一种表示形式是使用路线图来完成的。路线图是建立在通过边连接的节点上的图。每个节点代表机器人的一种可能状态，而每条边指示如何从另一个状态到达该状态。路线图的示例包括 Voronoi 图 [25]（见图 2e）、可见性图 [26] 和状态格图（见图 2f）。后者包括根据运动基元制作边缘，因此在给定机器人移动性约束的情况下，确保生成的路径是可行的，特别是在使用 *Graph Search* 算法时，如 Likhachev 和 Ferguson [27]、Bergman 等人的工作。 [28]。

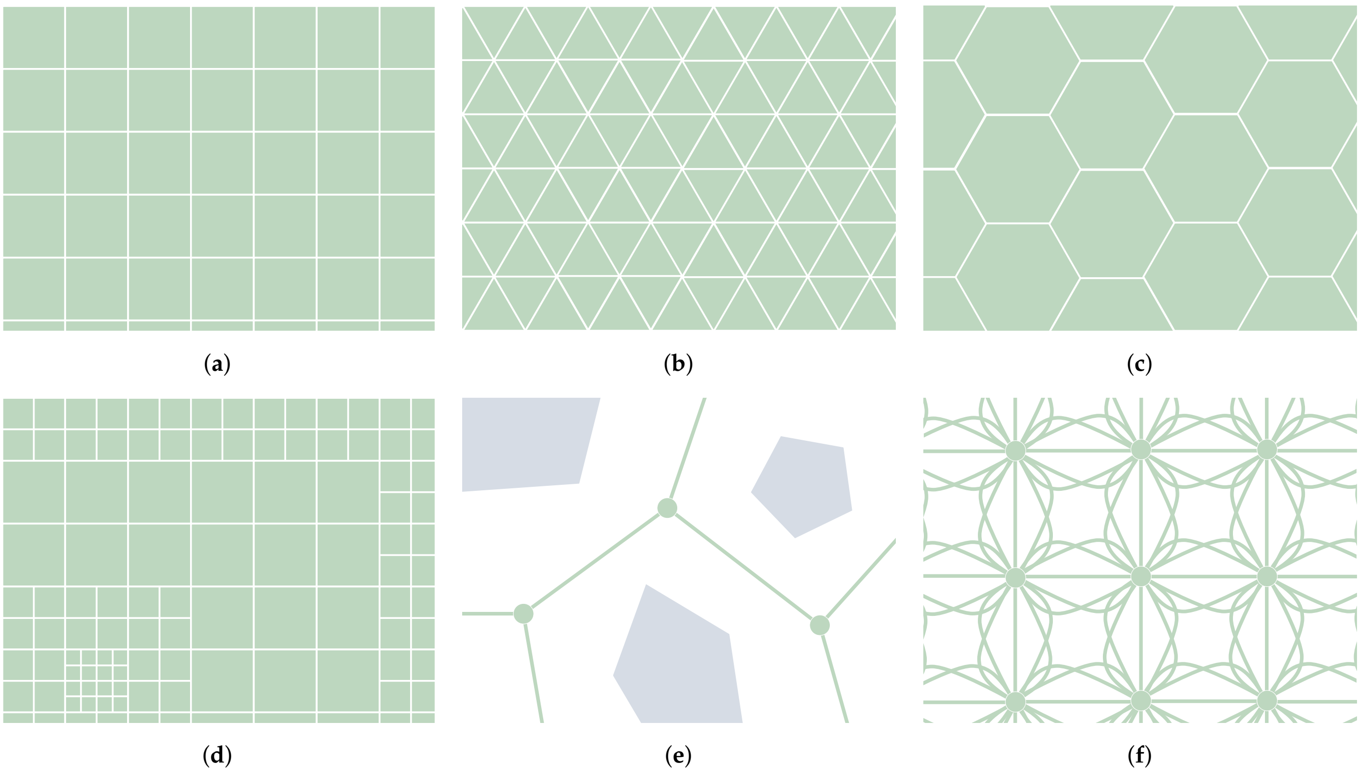


图 2. 不同类型的环境单元分解 (a–d) 和路线图 (e–f)。 (a) 使用正方形进行镶嵌。 (b) 使用三角形进行镶嵌。 (c) 使用六边形进行镶嵌。 (d) 不规则网格。 (e) 沃罗诺伊路线图。 (f) 状态格图。

这些图中的单元或节点可以以静态或动态元素的形式存储有关其所在位置的表面的信息[10]。例如，这可以是海拔信息。数字高程图 (DEM) 是一个网格，其中的每个节点都与一个高程值相关联。高程图也可以用多边形表示，但首选规则网格图[29]。可以使用卷积矩阵提取与形状相关的特征，例如斜率或表面粗糙度[30]。内核的大小和 DEM 分辨率将决定提取什么样的特征。此外，该分辨率定义了地图中包含的元素的详细程度。如图 2d 所示，该分辨率可以是不均匀的，也可以是多重的。网格的大小可以根据执行规划的规模来选择：*Local*表示覆盖机器人的直接周围环境（或多或少是机载传感器的可到达距离），而 *Global*表示面积大于此范围，通常使用来自卫星或无人机等外部来源的信息。

关于如何在规划器工作区定义成本，有不同的方法。首先，我们将成本理解为机器人通过移动积累的度量。路径规划器的目标是通过生成最佳路径来最小化这种累积。所讨论的成本可以是统一的，因为可以访问的区域

by the robot always have the same value. This approach can be used for collision-avoidance path planning, in which metrics such as the path length in a 2D plane are minimized. Non-uniform cost maps can be used to assign different values of cost to different accessible areas. It can be useful to, for example, define the energetic performance of the robot at each location. Moreover, the cost can be also defined according to a direction vector. This means that the robot will experience different values of cost depending on its heading. In this case, the cost is categorized as *anisotropic* [31], whereas in the contrary case the cost is *isotropic*. Furthermore, the steering manoeuvre of the robot can also have different values of cost according to its locomotion. Finally, it is worth noting that the environment can be fully known, partially known or even fully unknown, requiring for the latter two a planning strategy that is capable of replanning when this knowledge is updated.

2.2.2. Robot–Surface Interaction Modeling

A ground mobile robot interacts with the surface beneath it to propel itself. To perform this function, there are many different locomotion actuators, such as wheels, tracks, legs and even omnidirectional wheels. Figure 3 depicts three real examples of ground mobile robots using different configurations of actuators. These actuators, together with the joints linking them to the robot body, determine the kinematic structure and the dynamic behavior of the robot. In other words, they determine the locomotion configuration of the robot. Zhang et al. [20] summarize some kinematic and dynamic models of different well-known configurations: *Differential drive* [32] (see *Koguma* robot [33] in Figure 3a as an example and the depiction of the model in Figure 4a), *Ackermann steering* [34] (see Figure 4b,d), *Skid steering* [35] (see Figure 4c) and *Omnidirectional* [36]. Some of them entail constraints relevant to path planning, such as the minimum turning radius of robots with *Front Ackermann steering* [37] (see Figure 4b) or the high energy consumption of *Skid-steering* robots in turning manoeuvres [38]. Moreover, another model exists, called *Crabbing* (see Figure 4e). It allows a robot to drive in a direction different to the one it is facing, due to having steering joints on top of all wheels [39]. Furthermore, some kinematic configurations allow the robot to perform the *Point Turn* manoeuvre, which makes them rotate without translating. It is worth mentioning the existence of articulated robots that are capable of reconfiguring themselves to obtain some kind of benefit and perform multiple types of locomotion (see *SherpaTT* in Figure 3c as an example of this). For instance, articulated robots with tracks can actively control their stability while driving on rough terrains [40,41]. Others use a wheel-on-legs configuration to execute a locomotion mode called *Wheel-walking* [42,43]. This mode is designed to overcome soft terrains in which a robot could get stuck. In a similar way, *Push-pull* locomotion imitates the motion of a caterpillar to increase traction in this kind of terrain [44–46]. Path planning algorithms acknowledging this reconfiguration capability are a must-have for this kind of robot, as they can find paths that take advantage of their high adaptability. In the case of global planning, Rohmer et al. [47] used a *Graph Search* algorithm, Dijkstra, to first produce a path and later, via simulation tools, evaluate which locomotion mode is better to drive each of its parts. We, the authors of this review publication, proposed the use of a *PDE Solving* method to consider the multiple locomotion modes at the time of planning using an isotropic cost function [48,49]. To the authors' knowledge there are not many existing *Local Planning* approaches addressing the kinodynamic constraints of robots with multiple locomotion modes. Reid et al. [50] proposed the use of a *Sampling-Based* algorithm, the Fast Marching Tree (FMT*) , to tackle the motion planning of a reconfigurable hybrid robot with wheeled-legs.

The robot's locomotion will adapt better or worse depending on the terrain features, which were briefly mentioned before. These features may be related to either the morphology (shape) or the composition of the terrain. One of them is the terrain inclination or slope [51]. The slope has an influence on the *Roll* and *Pitch* orientation angles of the robot, which is important to consider to preserve stability [41,52,53]. Moreover, it can also influence the energetic performance of the robot according to its direction. This dependency on direction is due to the effect of gravity, making the robot consume different amounts of

机器人的值始终具有相同的值。这种方法可用于避免碰撞的路径规划，其中诸如 2D 平面中的路径长度等度量被最小化。非统一成本地图可用于将不同的成本值分配给不同的可访问区域。例如，它对于定义机器人在每个位置的能量性能很有用。此外，成本也可以根据方向向量来定义。这意味着机器人将根据其航向经历不同的成本值。在这种情况下，成本被分类为*anisotropic* [31]，而在相反的情况下，成本为*isotropic*。此外，机器人的转向操纵也可以根据其运动而具有不同的成本值。最后，值得注意的是，环境可以是完全已知的、部分已知的甚至完全未知的，后两者需要能够在知识更新时重新规划的规划策略。

2.2.2.机器人与表面交互建模

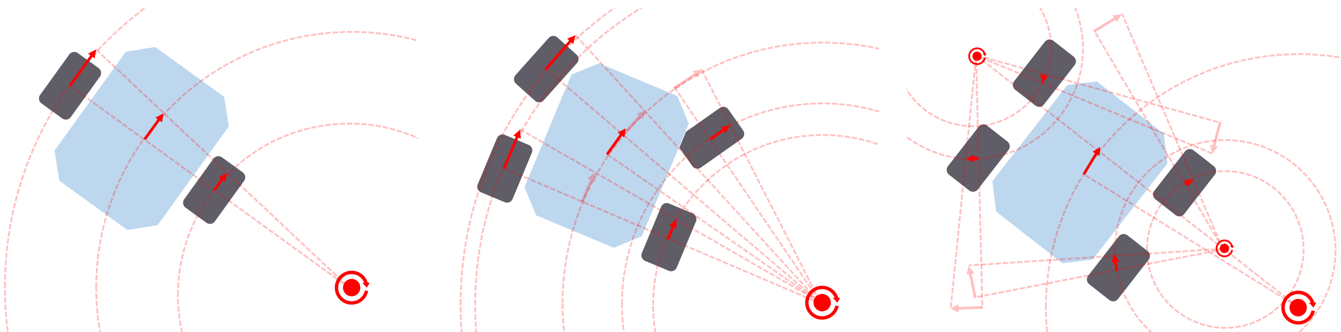
地面移动机器人与其下方的表面相互作用以推动自身。为了执行此功能，有许多不同的运动执行器，例如轮子、履带、腿甚至全向轮。图 3 描述了使用不同执行器配置的地面移动机器人的三个真实示例。这些执行器以及将它们连接到机器人主体的关节决定了机器人的运动结构和动态行为。换句话说，它们决定了机器人的运动配置。张等人。 [20]总结了不同众所周知的配置的一些运动学和动态模型：*Differential drive* [32]（参见图3a中的*Koguma*机器人[33]作为示例以及图4a中模型的描述），*Ackermann steering* [34]（参见图4b，d），*Skid steering* [35]（参见图4c）和*Omnidirectional* [36]。其中一些需要与路径规划相关的约束，例如*Front Ackermann steering*机器人的最小转弯半径[37]（见图4b）或*Skid-steering*机器人在转弯机动中的高能耗[38]。此外，还存在另一种模型，称为*Crabbing* (参见图 4e)。由于所有轮子顶部都有转向关节，它允许机器人沿着与其面对的方向不同的方向行驶[39]。此外，一些运动学配置允许机器人执行 *Point Turn* 机动，这使得它们旋转而不平移。值得一提的是，铰接式机器人的存在能够重新配置自身以获得某种利益并执行多种类型的运动（参见图 3c 中的 *SherpaTT* 作为示例）。例如，带履带的关节式机器人可以在崎岖地形上行驶时主动控制其稳定性[40,41]。其他人使用轮腿配置来执行称为 *Wheel-walking* [42,43] 的运动模式。该模式旨在克服机器人可能被卡住的松软地形。以类似的方式，*Push-pull* 运动模仿毛毛虫的运动，以增加这种地形的牵引力 [44-46]。承认这种重新配置能力的路径规划算法是此类机器人的必备条件，因为它们可以找到利用其高适应性的路径。就全球规划而言，Rohmer 等人。 [47] 使用 *Graph Search* 算法 Dijkstra 首先生成一条路径，然后通过模拟工具评估哪种运动模式更适合驱动其每个部分。我们，这篇评论出版物的作者，建议使用 *PDE Solving* 方法在使用各向同性成本函数进行规划时考虑多种运动模式 [48,49]。据作者所知，解决具有多种运动模式的机器人的运动动力学约束的现有 *Local Planning* 方法并不多。里德等人。 [50] 提出使用 *Sampling-Based* 算法，即快速行进树 (FMT*)，来解决带轮腿的可重构混合机器人的运动规划。

机器人的运动会根据地形特征而更好或更差地适应，这在前面已经简要提到过。这些特征可能与地形的形态（形状）或组成有关。其中之一是地形倾斜度或坡度[51]。斜率对机器人的 *Roll* 和 *Pitch* 方向角有影响，这对于保持稳定性很重要[41,52,53]。此外，它还可以根据机器人的方向影响其能量表现。这种对方向的依赖是由于重力的影响，使得机器人消耗不同数量的能量。

energy according to whether it is climbing, descending, going laterally or going diagonally through the slope [29,54–57]. Another relevant terrain feature is the roughness. This is the measure of how diverse the normal vectors are [58,59] and may affect the vibration experienced by the robot. Other terrain shapes can be negotiated by the robot, according to its chassis. For instance, it can overcome rocks using the body clearance, which is the space between the body lower surface and the terrain surface [60,61]. The presence of negative obstacles such as holes or ditches can be problematic as they are difficult to capture by the robot's sensors [62]. With regards to the terrain composition, it has an influence on the dynamics underlying the robot–surface interaction. This surface may be more rigid or deformable [63]. This affects the way the robot adheres to the surface, even restricting its motion [64]. The slippage is the metric that, in general terms, measures how the real speed of the robot differs from the commanded one, usually by calculating a ratio between them [65,66]. Some works consider both slippage and the slope of the terrain to make a more accurate estimation of the robot while traversing rough terrains [67,68]. The magnitude of gravity can also affect the dynamics of the interaction between the robot and the terrain [30,69]. Finally, path planners usually make use of cost functions that encompasses multiple terrain features related to shape and composition. For instance, Ishigami et al. [2] introduced the *dynamic mobility index*, encompassing stability, slippage, elapsed time and energy consumption. Moreover, there may exist other elements not directly related to the terrain that may still affect the performance of the robot while navigating. One of them is the solar radiation [66,70], which can be modeled as a dynamic function [71]. Groves et al. [72] mapped other kinds of radiation that may harm the robot, as in scenarios of nuclear dismantlement. Moreover, the concept of risk can be of high importance to prevent the robot from getting into a dangerous situation [73], such as increasing the cost with the proximity to obstacles [74].

(**a**) (**b**) (**c**)

Figure 3. Examples of ground mobile robots with different kinematic configurations. The *Koguma* rover (**a**) has *Differential Drive* locomotion. *Cuádriga* (**b**) is a *Skid-steering* robot owned by our institution, the University of Málaga. *SherpaTT* (**c**) has four steerable wheels that allow it to execute *Full Ackermann*, *Crabbing* and *Point Turn* maneuvers. (**a**,**c**) have been reproduced with permission of the University of Tohoku and the German Research Center for Artificial Intelligence (DFKI), respectively.



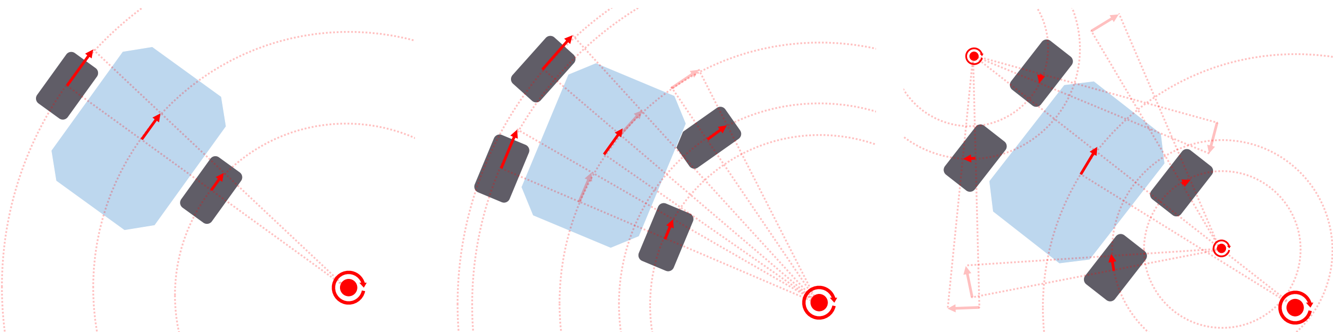
(**a**) (**b**) (**c**)

Figure 4. *Cont.*

能量取决于它是攀爬、下降、横向还是对角线穿过斜坡[29,54–57]。另一个相关的地形特征是粗糙度。这是法向向量多样性的度量[58,59]，并且可能会影响机器人所经历的振动。机器人可以根据其底盘协商其他地形形状。例如，它可以利用身体间隙克服岩石，身体间隙是身体下表面和地形表面之间的空间[60,61]。诸如洞或沟渠之类的负面障碍物的存在可能会产生问题，因为它们很难被机器人的传感器捕获[62]。就地形组成而言，它对机器人与表面相互作用的动力学有影响。该表面可能更坚硬或更易变形[63]。这会影响机器人粘附在表面上的方式，甚至限制其运动[64]。一般来说，滑移是衡量机器人实际速度与命令速度之间差异的指标，通常通过计算它们之间的比率[65,66]。一些工作考虑了滑移和地形的坡度，以便在穿越崎岖地形时对机器人进行更准确的估计[67,68]。重力的大小也会影响机器人与地形之间相互作用的动力学[30,69]。最后，路径规划者通常利用包含与形状和组成相关的多个地形特征的成本函数。例如，石上等人。 [2]引入了*dynamic mobility index*，包括稳定性、滑点、经过时间和能量消耗。此外，可能存在与地形不直接相关的其他因素，但仍可能影响机器人在导航时的性能。其中之一是太阳辐射[66,70]，它可以建模为动态函数[71]。格罗夫斯等人。 [72]绘制了可能伤害机器人的其他类型的辐射，例如在核拆除场景中。此外，风险的概念对于防止机器人陷入危险情况非常重要[73]，例如靠近障碍物会增加成本[74]。

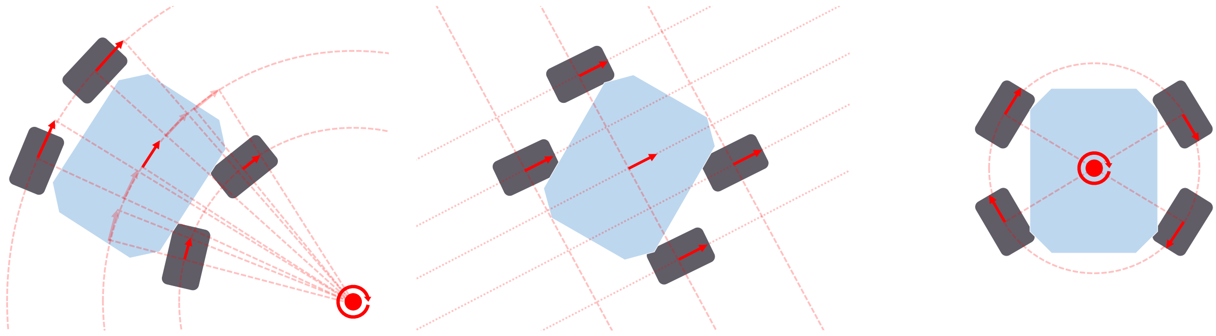
(**a**) (**b**) (**c**)

图 3.具有不同运动学配置的地面移动机器人示例。 *Koguma* 漫游车 (a) 具有 *Differential Drive* 运动。 *Cuádriga* (b) 是我们马拉加大学拥有的一个 *Skid-steering* 机器人。 *SherpaTT* (c) 有四个可转向轮，使其能够执行 *Full Ackermann*、*Crabbing* 和 *Point Turn* 机动。 (a,c) 的复制已分别获得东北大学和德国人工智能研究中心 (DFKI) 的许可。



(**a**) (**b**) (**c**)

Figure 4. *Cont.*



(**d**) (**e**) (**f**)

Figure 4. Locomotion models used for wheeled ground mobile robots along with path planners: Differential drive (**a**), Front Ackermann (**b**), Skid-steering (**c**), Full Ackermann (**d**), Crabbing (**e**) and Point Turn (**f**).

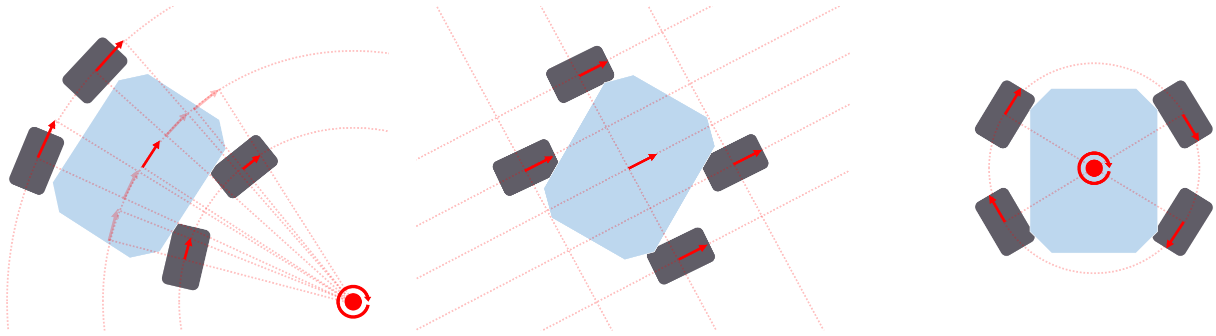
3. Reactive-Computing-Based Path Planning Algorithms

This category encompasses path planning algorithms where the environment, usually a map distinguishing between obstacle and non-obstacle regions, only indicates the location and shape of the existing obstacles. *Reactive Computing* algorithms are usually employed as local path planners (covering the surroundings of the robot and with dynamic replanning) due to their capability to quickly handle new information (e.g., in the form of newly discovered obstacles), which often comes from the limited onboard sensors. As local planners, these algorithms usually plan the next immediate path or manoeuvre to avoid nearby obstacles while following a global plan made by another algorithm. However these algorithms may calculate local minima paths, or even cause the robot to get stuck, so special care must be taken. There are two subcategories of *Reactive Computing* algorithms: *Reactive Manoeuvre* methods, where the presence of obstacles determines the immediate next manoeuvre of the robot, and *Local Optimization* methods, where an existing path is modified according to the presence of obstacles.

3.1. Reactive Manoeuvre

The algorithms presented here rely on defining how the robot reacts at each instant to the presence of obstacles. This reaction can be defined according to a formulation that addresses the location of existing obstacles. A common feature of the different formulation approaches is the low computational requirements needed to produce the reaction, usually in the form of a steering or velocity command. Since this formulation lacks global information, these techniques are commonly used as *Local Planners*. The formulation in question may be based on the use of fields to tackle the location of obstacles, the detection of obstacle boundaries to circumvent them, or the production of a velocity command after evaluating the available free space or the speed of moving obstacles.

The use of field methods include the Artificial Potential Fields (APF) and the Vector Field Histogram (VFH) algorithms. In APF, the motion of a robot can result from the sum of virtual forces that external elements such as obstacles create. In this way, the robot gets further from these obstacles and avoids colliding with them, as the forces from them are repulsive [75]. An attractive force created by the target position makes the robot go towards it. This can be seen in the picture displayed in Figure 5a. Ge and Cui [76] presented an application of APF in environments containing dynamic obstacles. However, the main drawback of this strategy is that it is prone to causing the robot to get stuck in local minimum points. For this reason, further research works were oriented to overcome this issue. This was the main target of the work by Vadakkepat et al. [77]. They combined APF with genetic methods (*Evolutionary* algorithms) to overcome this situation. This combination was also used to plan the motion of a simulated 6-wheel rover [78]. Another hybrid version includes the use of PSO algorithms [79]. Triharminto et al. [80] proposed a non-hybrid solution, consisting of adding a repulsive potential field around the rover. However, it has not yet been tested with U-shaped obstacles. Other works solve this



(**d**) (**e**) (**f**)

图 4. 用于轮式地面移动机器人和路径规划器的运动模型：差速驱动 (a)、前阿克曼 (b)、滑移转向 (c)、完全阿克曼 (d)、爬行 (e) 和点转弯 (f)。

3. 基于反应计算的路径规划算法

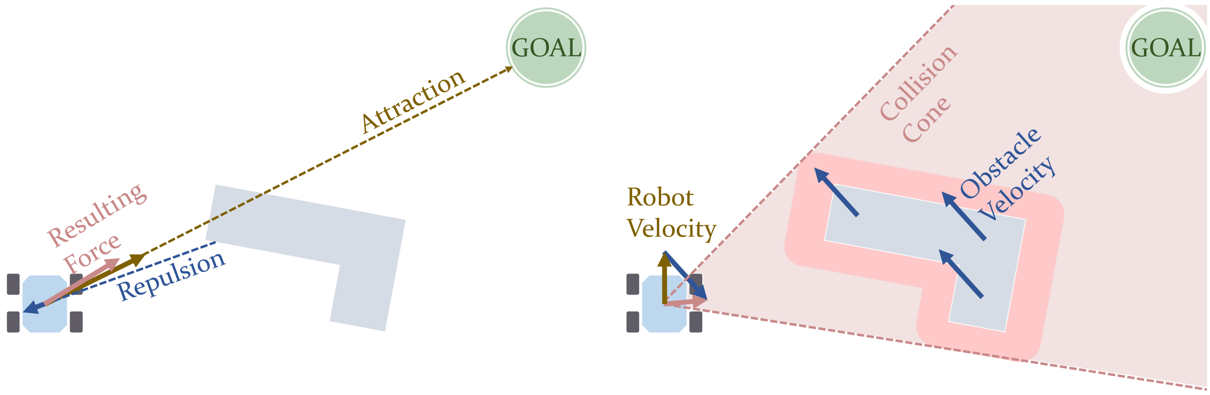
此类别包含路径规划算法，其中环境（通常是区分障碍物和非障碍物区域的地图）仅指示现有障碍物的位置和形状。 *Reactive Computing*算法通常被用作本地路径规划器（覆盖机器人的周围环境并进行动态重新规划），因为它们能够快速处理新信息（例如，以新发现的障碍物的形式），而新信息通常来自有限的机载传感器。作为局部规划器，这些算法通常会规划下一个直接路径或机动以避免附近的障碍物，同时遵循另一个算法制定的全局计划。然而这些算法可能会计算出局部极小路径，甚至导致机器人卡住，所以必须特别小心。 *Reactive Computing* 算法有两个子类别：*Reactive Manoeuvre* 方法，其中障碍物的存在决定机器人的下一个动作；以及 *Local Optimization* 方法，其中根据障碍物的存在修改现有路径。

3.1. Reactive Manoeuvre

这里介绍的算法依赖于定义机器人在每个时刻对障碍物的存在如何反应。这种反应可以根据解决现有障碍物位置的公式来定义。不同公式方法的一个共同特点是产生反应所需的计算要求较低，通常以转向或速度命令的形式。由于该公式缺乏全局信息，因此这些技术通常用作 *Local Planners*。所讨论的公式可能基于使用场来解决障碍物的位置、检测障碍物边界以绕过障碍物、或者在评估可用自由空间或移动障碍物的速度后生成速度命令。

场方法的使用包括人工势场 (APF) 和矢量场直方图 (VFH) 算法。在 APF 中，机器人的运动可能是由障碍物等外部元素产生的虚拟力的总和产生的。通过这种方式，机器人可以远离这些障碍物并避免与它们碰撞，因为它们的力是排斥的[75]。目标位置产生的吸引力使机器人朝目标位置移动。这可以从图 5a 中显示的图片中看出。 Ge 和 Cui [76] 提出了 APF 在包含动态障碍物的环境中的应用。然而，该策略的主要缺点是容易导致机器人陷入局部极小点。因此，进一步的研究工作旨在解决这个问题。这是 Vadakkepat 等人工作的主要目标。 [77]。他们将 APF 与遗传方法（*Evolutionary* 算法）相结合来克服这种情况。该组合还用于规划模拟 6 轮漫游车的运动 [78]。另一个混合版本包括 PSO 算法的使用 [79]。 Triharminto 等人。 [80]提出了一种非混合解决方案，包括在流动站周围添加一个排斥势场。不过，它尚未经过 U 形障碍物的测试。其他作品解决了这个问题

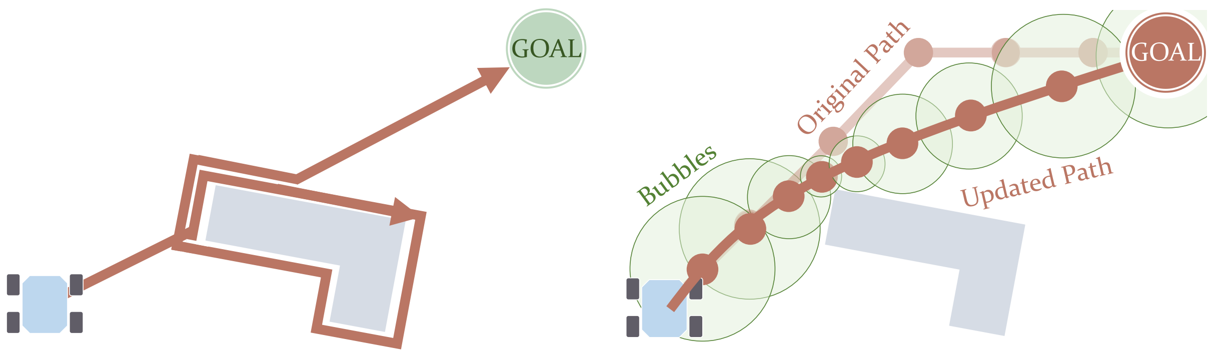
issue by creating custom escape paths whenever it is detected that the robot is on a local minimum point [81]. Furthermore, Bayat et al. [82] present a solution inspired by electrostatic potential fields in which, instead of using a sum of virtual forces, a so-called scalar potential field guides the robot. VFH, proposed by Borenstein et al. [83], creates a polar histogram to evaluate the density of obstacles around the robot, selecting the steering angle with the lowest density of obstacles. Some improvements to VFH were introduced years later [84,85].



(**a**) (**b**)

Figure 5. Graphical representations of concepts used in Artificial Potential Field (**a**) and Velocity Obstacle (**b**) algorithms. (**a**) Potential fields acting on a robot. (**b**) Collision cone considering a moving obstacle.

The Bug algorithms, Bug1 and Bug2, make the robot circumvent any obstacle found on its way until it reaches the goal [86]. The main difference between them is that Bug1 makes the robot drive the full boundary of any obstacle (see Figure 6a), whereas Bug2 can drive it only partially [19]. They despise optimality in favour of easiness in the implementation and very minimal computation. They can be used on robots equipped only with sensors that just detect obstacles in their immediate vicinity. In this way, these robots either drive towards the goal or drive along the boundaries of obstacles they find. The works of Buniyamin et al. [87] and Campbell et al. [19] refer to some variants that improve this kind of algorithm, in general reducing the distance the robot drives. Xu et al. [88] uses the Bug algorithm considering turning radius constraints, producing paths with smooth turns.

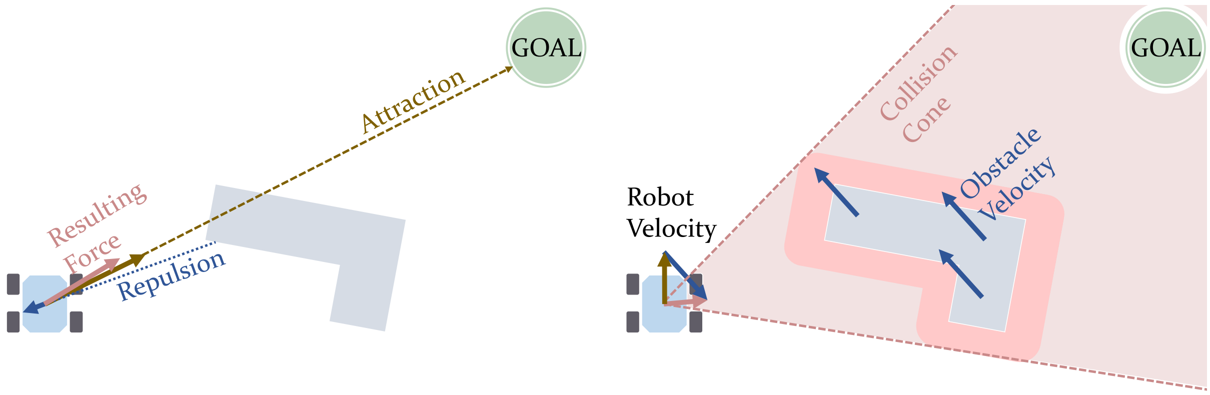


(**a**) (**b**)

Figure 6. Graphical representations of concepts used in the Bug algorithm (**a**) and Elastic Bands (**b**) algorithm with Bubble bounds. (**a**) Path after using the Bug1 algorithm. (**b**) Path after experiencing stretching.

The following approaches are focused on producing a velocity command for the robot. Velocity Obstacles methods calculate a safe trajectory considering the velocity vectors of both the robotic agent and any other moving obstacle [89,90]. This calculation evaluates a cone of collision such as the one shown in Figure 5b. Chen et al. [91] used Velocity Obstacles in an hybrid fashion with another algorithms called Fast Marching

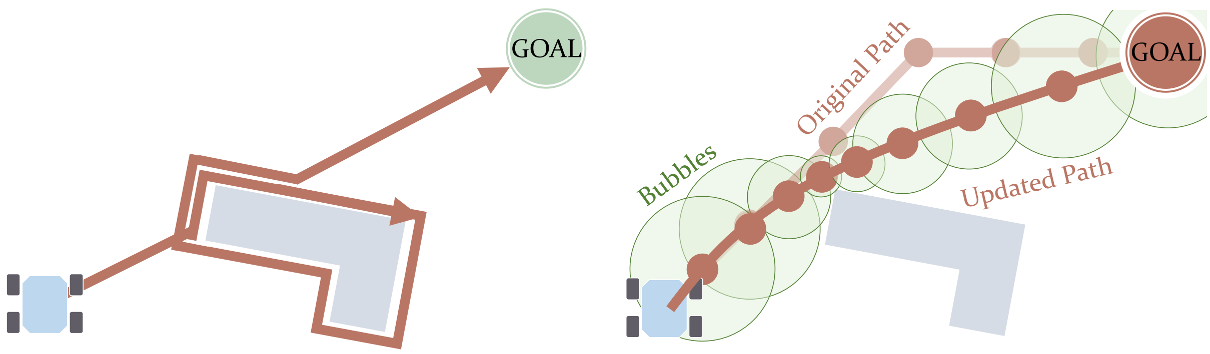
每当检测到机器人位于局部最小值点时，都会通过创建自定义逃生路径来解决问题[81]。此外，巴亚特等人。 [82]提出了一种受静电势场启发的解决方案，其中不使用虚拟力的总和，而是使用所谓的标量势场引导机器人。 VFH，由 Borenstein 等人提出。 [83]，创建极坐标直方图来评估机器人周围障碍物的密度，选择障碍物密度最低的转向角度。几年后对 VFH 进行了一些改进[84,85]。



(**a**) (**b**)

图 5. 人工势场 (a) 和速度障碍 (b) 算法中使用的概念的图形表示。 (a) 作用在机器人上的势场。 (b) 考虑移动障碍物的碰撞锥
。

Bug 算法 Bug1 和 Bug2 使机器人绕过途中发现的任何障碍物，直到到达目标 [86]。它们之间的主要区别在于，Bug1 使机器人能够驱动任何障碍物的完整边界（见图 6a），而 Bug2 只能部分驱动机器人 [19]。他们鄙视最优性，而倾向于实现的简便性和极少的计算量。它们可用于仅配备传感器的机器人，这些传感器仅检测附近的障碍物。通过这种方式，这些机器人要么朝着目标行驶，要么沿着它们发现的障碍物边界行驶。布尼亚明等人的作品。 [87] 和坎贝尔等人。 [19]提到了改进此类算法的一些变体，总体上减少了机器人行驶的距离。徐等人。 [88]使用考虑转弯半径约束的Bug算法，产生平滑转弯的路径。



(**a**) (**b**)

图 6. Bug 算法 (a) 和带气泡边界的弹性带 (b) 算法中使用的概念的图形表示。 (a) 使用Bug1算法后的路径。 (b) 经历拉伸后的路径。

以下方法侧重于为机器人生成速度命令。速度障碍方法考虑机器人代理和任何其他移动障碍物的速度矢量来计算安全轨迹[89,90]。该计算评估如图 5b 所示的碰撞锥体。陈等人。 [91] 以混合方式使用 Velocity Obstacles 和另一种称为 Fast Marching 的算法

Square (FMS). Wilkie et al. [92] considered the *Front-ackermann* (see Figure 4b) locomotion model for cars. Chakravarthy and Ghose [93] proposed using a Collision-cone algorithm, similar to Velocity Obstacles but considering obstacles with arbitrary shapes. Qu et al. [94] introduced its use along with car models (with *Front-ackermann* configurations, as seen in Figure 4b). Finally, the Dynamic Window Approach (DWA) is an algorithm that searches in the velocity space for a velocity command to follow a collision-free circular trajectory, delimited by admissible speed values and a time window [95]. This solution may not be the globally optimal one, but rather a locally optimal one [4]. DWA can be used even for robots navigating at high speeds [96]. Feng et al. [97] propose an improved version that reduces its complexity by reducing the velocity space to search. Other approaches use this algorithm for energy-minimization path planning [98,99]. Although commonly used as a local planner, Zhang et al. [5] have proposed its use at a global scale as a global path planner.

3.2. Local Optimization

These algorithms usually start from a pre-existing path and modify it according to the existing obstacles. Here it is prioritized to keep computational use to the minimum at the expense of losing optimality or even completeness. There are different options to modify the path, ranging from the selection of velocity profiles within a velocity space to the stretching and elongation of the path under the effect of artificial forces.

The use of Elastic Bands in path planning was proposed by Quinlan and Khatib [100]. This method deforms an existing collision-free path according to the obstacles, stretching (see Figure 6b) or elongating it. From a set of points in this path, a set of overlapping subregions, called Bubbles, is created. These Bubbles cover collision-free areas and their size is bounded by the distance to obstacles. This entails that smaller and more numerous Bubbles are present in the portions of the path closer to obstacles. It can be used in dynamic environments, although large changes may lead to the failure of the method [100]. Moreover, it has also been adapted to non-holonomic vehicles by complying with curvature constraints Khatib et al. [101] and using *Bezier* curves [102]. An extension to Elastic Bands includes time constraints and is named Timed Elastic Bands (TEB). This extended version addresses the kinodynamic constraints of the robot [103].

4. Soft-Computing-Based Path Planning Algorithms

This kind of algorithm does not intend to find the exact optimal solution, but rather to approximate it, tolerating a certain range of imprecision. In general, these algorithms require the tuning of certain parameters by the user in order to work properly according to the characteristics of the environment. They can deal even with dynamic environments and are adequate for problems involving a large number of variables and high degrees of freedom [8]. However, in general they demand a high number of computational resources. This review follows the classification proposed by Mirjalili and Dong [104], which distinguishes between *Evolutionary*, *Fuzzy control* and *Machine learning* methods. The first one uses techniques inspired by biology and nature: they start with a system formed by individuals that changes over time, i.e., evolves. *Fuzzy control* and *Machine learning* methods are here part of a subcategory named *Artificial Intelligence*. They use fuzzy rules and neural networks, respectively, to produce controllers. These controllers are very useful for navigating through initially unknown scenarios and in general produce paths according to the obstacles the robot detects on its way. To sum up, *Soft Computing* algorithms allow the tuning of a series of repetitive elements, either nature-based individuals, fuzzy rules or artificial neurons, to generate a path.

4.1. Evolutionary Computation

Evolutionary algorithms are also known as *Meta-heuristic* or Nature-inspired [105]. These algorithms generate a path that results from the evolution of a population. This population is made up of intelligent individuals whose actions are modelled after behaviors

广场（FMS）。威尔基等人。 [92]考虑了汽车的*Front-ackermann* (参见图 4b) 运动模型。 Chakravarthy 和 Ghose [93] 提出使用碰撞锥算法，类似于速度障碍，但考虑任意形状的障碍物。曲等人。 [94]介绍了它与汽车模型的使用（具有 *Front-ackermann* 配置，如图 4b 所示）。最后，动态窗口方法（DWA）是一种在速度空间中搜索速度命令以遵循无碰撞圆形轨迹的算法，该轨迹由允许的速度值和时间窗口界定[95]。该解决方案可能不是全局最优的，而是局部最优的[4]。 DWA 甚至可以用于高速导航的机器人 [96]。冯等人。 [97]提出了一种改进的版本，通过减少搜索的速度空间来降低其复杂性。其他方法使用该算法进行能量最小化路径规划[98,99]。尽管通常被用作当地规划师，但张等人。 [5]建议在全球范围内使用它作为全球路径规划器。

3.2. Local Optimization

这些算法通常从预先存在的路径开始，并根据现有的障碍对其进行修改。在这里，优先考虑将计算使用量保持在最低限度，但代价是失去最优性甚至完整性。修改路径有不同的选项，从速度空间内的速度剖面的选择到在人为力的作用下拉伸和伸长路径。

Quinlan 和 Khatib [100] 提出在路径规划中使用弹性带。该方法根据障碍物使现有的无碰撞路径变形，拉伸（见图6b）或拉长它。从该路径中的一组点，创建一组重叠的子区域，称为气泡。这些气泡覆盖无碰撞区域，其大小受距障碍物的距离限制。这意味着在靠近障碍物的路径部分中存在更小和更多的气泡。它可以在动态环境中使用，尽管较大的变化可能会导致该方法的失败[100]。此外，它还通过遵守曲率约束 Khatib 等人，适用于非完整车辆。 [101]并使用*Bezier*曲线[102]。弹性带的扩展包括时间限制，称为定时弹性带 (TEB)。这个扩展版本解决了机器人的运动动力学约束[103]。

4. 基于软计算的路径规划算法

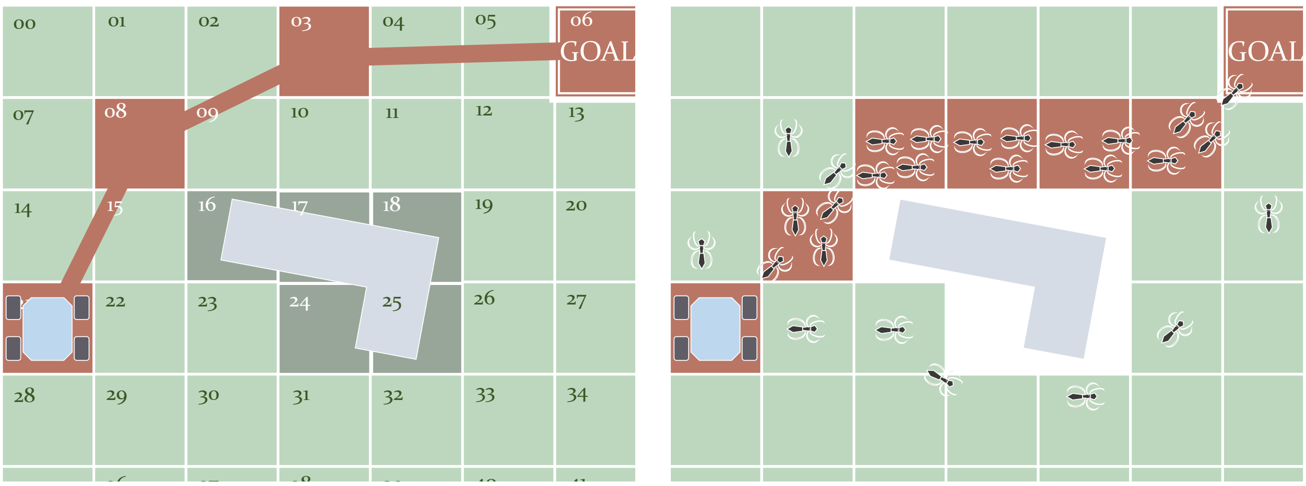
这种算法并不打算找到精确的最优解，而是近似它，容忍一定范围的不精确性。一般来说，这些算法需要用户调整某些参数才能根据环境的特点正常工作。它们甚至可以处理动态环境，并且足以解决涉及大量变量和高自由度的问题[8]。然而，通常它们需要大量的计算资源。本综述遵循 Mirjalili 和 Dong [104] 提出的分类，该分类区分了 *Evolutionary*、*Fuzzy control* 和 *Machine learning* 方法。第一个使用受生物学和自然启发的技术：它们从由个体形成的系统开始，该系统随着时间的推移而变化，即进化。 *Fuzzy control* 和 *Machine learning* 方法是名为 *Artificial Intelligence* 的子类别的一部分。他们分别使用模糊规则和神经网络来生成控制器。这些控制器对于在最初未知的场景中导航非常有用，并且通常根据机器人在途中检测到的障碍物生成路径。总而言之，*Soft Computing*算法允许调整一系列重复元素（基于自然的个体、模糊规则或人工神经元）来生成路径。

4.1. Evolutionary Computation

Evolutionary 算法也称为*Meta-heuristic*或自然启发的[105]。这些算法生成一条由种群进化产生的路径。这个群体由聪明的个体组成，他们的行为是模仿行为的

found in nature [104]. These actions may involve modifying themselves and/or interacting with other individuals. In some cases, these operations imply a motion by the individuals in the free space of the environment, i.e., in the space reachable by the robot. After performing a series of these operations, the algorithms approximate the optimal solution. The resulting path and the time invested to converge depend on the behavior policy assigned to the individuals, the nature of the scenario and the values assigned by the user to certain configurable parameters. An example of the latter is the number of individuals that populate the path planning problem. *Evolutionary* algorithms include *Genetic* methods and *Swarm Optimizers*. The first method uses chromosome models, whereas the second one models the behavior of living beings.

As just mentioned, *Genetic* algorithms work with individuals modeled after chromosomes [106]. These individuals contain genes, as chromosomes do, in the form of binary numbers. These numbers encode a solution, which is the set of *waypoints* forming a path in the particular case of solving the path planning problem. In other words, each chromosome in the population represents a path. Figure 7a depicts an example using a grid. This grid is made up of cells, each of them labeled with a number. The genetic algorithm starts with a random set of chromosomes. This set evolves using three processes: *Reproduction*, *Crossover* and *Mutation* [107]. *Reproduction* creates new chromosomes by copying the best ones. It also removes the worst ones. *Crossover* is the process in which chromosomes interchange their genes. *Mutation* introduces random changes in the genes to incentive the exploring of the search space and to avoid local minima. As a result of continuously repeating these processes, the algorithm converges. Zhang et al. [4] mention *Genetic* algorithms that converge more slowly as they get closer to the optimal solution. Han et al. [108] used *Genetic* algorithms to find the shortest paths in environments with dynamic obstacles. Tuncer and Yildirim [109] presented a modification to the *Mutation* process. This consists of checking free nodes around the position that is about to be mutated. This work compares the results with previous variants of the method. Another research work used genetic algorithms, along with large grids up to 2000 $\times$ 2000 nodes [110]. Genetic algorithms for path planning were also tested on an experimental platform [111]. More improvements include the initial selection of *waypoints*, considering those located near obstacles [112]. Elhoseny et al. [113] introduced a more regulatory policy on exploration during *Mutation*, taking into account how diverse the chromosomes are. Furthermore, this work not only optimized path length but also preserved smoothness using *Bézier* curves. Finally, the *Crossover* process is also improved upon in the work of Lamini et al. [114] to make the solution converge faster and also to reduce the number of turns made by the robot. There are many path planning applications based on *Genetic* methods presented in the review of Patle et al. [10].

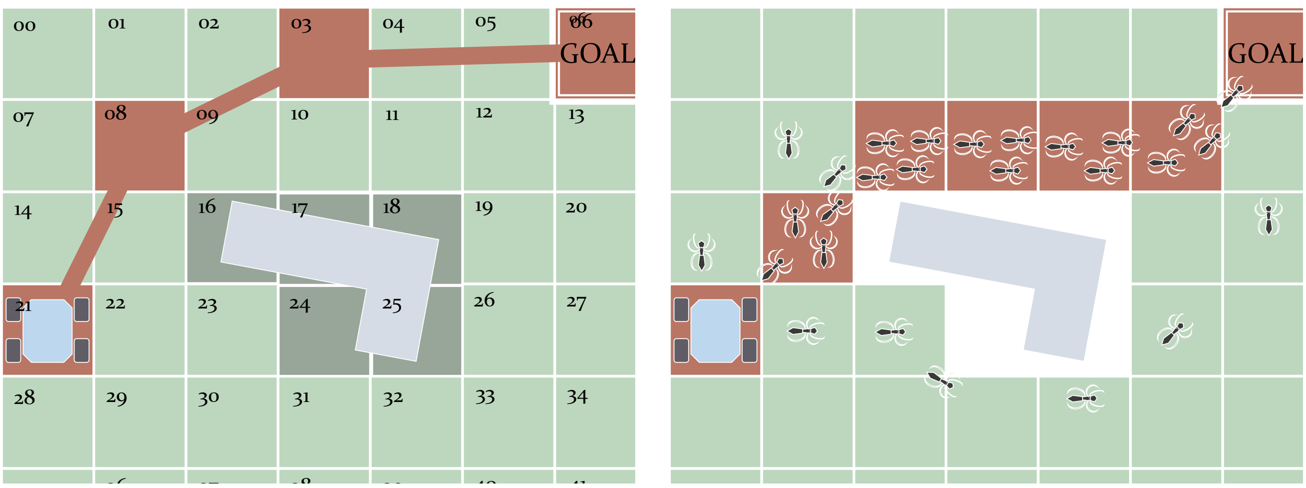


(**a**) (**b**)

Figure 7. Examples of *Evolutionary* algorithms: *Genetic* (**a**) and *Swarm Optimizer* ACO (**b**). (**a**) Functioning of the genetic algorithms used to perform path planning. The path in this figure has the form of a chromosome with genes 21–08–03–06. (**b**) Functioning of the ACO algorithm. Simulated ants deposit more pheromones in the shortest path. Eventually the majority will follow this path.

自然界中发现的[104]。这些行为可能涉及改变自己和/或与其他人互动。在某些情况下，这些操作意味着个体在环境的自由空间（即机器人可到达的空间）中的运动。执行一系列这些操作后，算法会逼近最优解。最终的路径和收敛时间取决于分配给个人的行为策略、场景的性质以及用户为某些可配置参数分配的值。后者的一个例子是路径规划问题中的个体数量。 *Evolutionary*算法包括*Genetic*方法和*Swarm Optimizers*。第一种方法使用染色体模型，而第二种方法则模拟生物的行为。

正如刚才提到的，*Genetic* 算法适用于以染色体为模型的个体[106]。这些个体含有二进制数形式的基因，就像染色体一样。这些数字编码一个解决方案，它是在解决路径规划问题的特定情况下形成路径的 *waypoints* 的集合。换句话说，群体中的每条染色体都代表一条路径。图 7a 描述了使用网格的示例。该网格由单元格组成，每个单元格都标有一个数字。遗传算法从一组随机染色体开始。该集合使用三个过程演化：*Reproduc-*、*tion*、*Crossover* 和 *Mutation* [107]。 *Reproduction* 通过复制最好的染色体来创建新的染色体。它还删除了最糟糕的。 *Crossover* 是染色体交换基因的过程。 *Mutation* 在基因中引入随机变化，以激励搜索空间的探索并避免局部最小值。由于不断重复这些过程，算法收敛。张等人。 [4]提到*Genetic*算法随着接近最优解而收敛得更慢。韩等人。 [108]使用*Genetic*算法来寻找具有动态障碍物的环境中的最短路径。 Tuncer 和 Yildirim [109] 提出了对 *Mutation* 过程的修改。这包括检查即将发生突变的位置周围的空闲节点。这项工作将结果与该方法以前的变体进行了比较。另一项研究工作使用遗传算法以及高达 2000 × 2000 个节点的大型网格[110]。用于路径规划的遗传算法也在实验平台上进行了测试[111]。更多改进包括考虑到位于障碍物附近的物体，对 *waypoints* 的初始选择[112]。埃尔霍森尼等人。 [113]在*Mutation*期间引入了更多的勘探监管政策，考虑到染色体的多样性。此外，这项工作不仅优化了路径长度，而且还使用 *Bézier* 曲线保持了平滑度。最后，Lamini 等人的工作中也改进了 *Crossover* 流程。 [114]使解决方案收敛得更快，并减少机器人的转弯次数。 Patle 等人的评论中提出了许多基于 *Genetic* 方法的路径规划应用程序。 [10]。



(**a**) (**b**)

图 7. *Evolutionary* 算法示例：*Genetic* (a) 和 *Swarm Optimizer* ACO (b)。 (a) 用于执行路径规划的遗传算法的功能。该图中的路径具有带有基因 21-08-03-06 的染色体的形式。 (b) ACO 算法的运行。模拟蚂蚁在最短的路径上沉积更多的信息素。最终大多数人都会走这条路。

Unlike the algorithms based on *Genetic* methods, *Swarm Optimizers* use agents that move and actuate in the free space. These individuals are modeled after animals in most cases. After a series of iterations, the motion of these individuals towards the goal creates a pattern that eventually converges to the resulting path. Table 2 introduces some of the models used that can be found in the literature. The Particle Swarm Optimization (PSO) algorithm stands out because of its simplicity. It is inspired by the behavior of certain groups of animals such as fish schools [115]. It creates a series of particles that relocate themselves over time until the algorithm converges. These algorithms search for the best positions and communicate with each other, considering their previous experience [116]. Mac et al. [117] propose a path planner that combines PSO with the Dijkstra algorithm (a *Graph Search* planner that is discussed below). Another well-known algorithm is the Ant Colony Optimizer (ACO) which, as the name indicates, simulates the behavior of ants. These insects move while leaving a trail of pheromones in their search for food. This trail can be tracked by the rest of the ants. Those places that contain more pheromones make up the *waypoints* of the best-found path. Figure 7b depicts this concept in a situation with an obstacle between the start and goal positions. Here, the best path is the shortest one. Following the same strategy, virtual ants can move on a grid, leaving more or fewer pheromones according to their state concerning the goal [118,119]. To avoid falling into a local minimum, some works combine this method with heuristic functions [120,121]. Che et al. [122] also solve this by adopting a rule used in the Grey Wolves approach. Luo et al. [123] present some improvements for ACO to not only avoid deadlocks but also to reduce the time it needs to converge. This algorithm has also been simulated with DEMs to minimize time [124] and energy [125]. The latter has also been achievedwhile avoiding dynamic obstacles, which is also addressed by the work of Sangeetha et al. [126]. This work combines ACO with fuzzy control. There are plenty more nature-inspired approaches. To avoid extending too much, some of them are indicated in Table 2. Moreover, there are also cases in which two models are combined. This is the case of the approach presented by Saraswathi et al. [127], where a hybrid between Cuckoo and Bat algorithms is tested.

Table 2. Intelligent models used for some swarm algorithms, together with some of the models.

Individuals	Behaviors
Particles [115–117,128]	Best position memory, variable velocity
Fireflies [129]	Brightness atraction
Dragonflies [130]	Hunting and migration
Grasshoppers [131,132]	Attraction and repulsion
Wolves [133,134]	Hierarchy system, hunting for prey
Whales [135,136]	Spiral bubble-nets
Cuckoos [137]	Lévy flights, brood parasitism
Bats [138]	Echolocation
Bacteria [139]	Foraging

4.2. Artificial Intelligence

Soft Computing algorithms may use other sets of configurable operators such as fuzzy rules or neural networks. Seraji and Howard [140] demonstrate the use of fuzzy logic on an experimental mobile platform to navigate through unstructured terrain. Zavlangas and Tzafestas [141] provided a fuzzy-logic-based system designed to make a mobile robot autonomously navigate through a dynamic environment, avoiding obstacles on its way. The work of Wang et al. [142] focuses on preventing the robot from getting stuck in local minimum points, such as those produced by U-shaped obstacles. Pandey et al. [143] also provides the results of simulation tests of fuzzy logic for obstacle avoidance. With regards to the use of neural networks, Yan and Li [144] also use fuzzy logic on a platform focusing on minimizing computational resources and traversing environments containing dynamic obstacles. Pandey and Parhi [145] combines fuzzy logic and a population-based algorithm named Wind Driven Optimization that tunes the fuzzy rules. With regards to the use

与基于 *Genetic* 方法的算法不同，*Swarm Optimizers* 使用在自由空间中移动和驱动的代理。在大多数情况下，这些个体都是以动物为模型的。经过一系列迭代后，这些个体朝着目标的运动创建了一种模式，最终收敛到最终的路径。表 2 介绍了一些可以在文献中找到的使用模型。粒子群优化 (PSO) 算法因其简单性而脱颖而出。它受到某些动物群体（例如鱼群）行为的启发[115]。它创建了一系列粒子，这些粒子会随着时间的推移重新定位，直到算法收敛。这些算法考虑到他们以前的经验，搜索最佳位置并相互通信[116]。麦克等人。 [117]提出了一种将 PSO 与 Dijkstra 算法相结合的路径规划器（下面讨论的 *Graph Search* 规划器）。另一个著名的算法是蚁群优化器（ACO），顾名思义，它模拟蚂蚁的行为。这些昆虫在寻找食物时会在移动时留下信息素的踪迹。其余的蚂蚁可以追踪这条踪迹。那些包含更多信息素的地方构成了最佳找到路径的*waypoints*。图 7b 描述了在起始位置和目标位置之间有障碍物的情况下的这一概念。在这里，最好的路径是最短的路径。遵循相同的策略，虚拟蚂蚁可以在网格上移动，根据它们关于目标的状态留下更多或更少的信息素[118,119]。为了避免陷入局部最小值，一些工作将此方法与启发式函数结合起来[120,121]。车等人。 [122]也通过采用灰狼方法中使用的规则来解决这个问题。罗等人。 [123]对 ACO 提出了一些改进，不仅可以避免死锁，还可以减少收敛所需的时间。该算法还使用 DEM 进行了模拟，以最大限度地减少时间 [124] 和能量 [125]。后者也在避免动态障碍的同时实现了，Sangeetha 等人的工作也解决了这个问题。 [126]。这项工作将 ACO 与模糊控制结合起来。还有更多受自然启发的方法。为了避免扩展过多，表2中列出了其中一些。此外，也存在两种模型组合的情况。 Saraswathi 等人提出的方法就是这种情况。 [127]，其中测试了 Cuckoo 和 Bat 算法的混合算法。

表 2. 用于一些群体算法的智能模型以及一些模型。

Individuals	**Behaviors**
Particles [115–117,128]	Best position memory, variable velocity
Fireflies [129]	Brightness atraction
Dragonflies [130]	Hunting and migration
Grasshoppers [131,132]	Attraction and repulsion
Wolves [133,134]	Hierarchy system, hunting for prey
Whales [135,136]	Spiral bubble-nets
Cuckoos [137]	Lévy flights, brood parasitism
Bats [138]	Echolocation
Bacteria [139]	Foraging

4.2. Artificial Intelligence

Soft Computing 算法可以使用其他可配置运算符集，例如模糊规则或神经网络。 Seraji 和 Howard [140] 演示了在实验移动平台上使用模糊逻辑来导航非结构化地形。 Zavlangas 和 Tzafestas [141] 提供了一个基于模糊逻辑的系统，旨在使移动机器人在动态环境中自主导航，避开途中的障碍物。王等人的工作。 [142]专注于防止机器人陷入局部极小点，例如 U 形障碍物产生的极小点。潘迪等人。 [143]还提供了模糊逻辑避障的模拟测试结果。关于神经网络的使用，Yan 和 Li [144] 也在一个平台上使用模糊逻辑，重点是最小化计算资源和穿越包含动态障碍物的环境。 Pandey 和 Parhi [145] 将模糊逻辑和名为“风驱动优化”的基于群体的算法相结合，用于调整模糊规则。关于使用

of neural networks, Zou et al. [146] presented a brief survey on this kind of algorithm for applications at the beginning of the 2000s. Engedy and Horváth [147] presented a path planner using neural networks for mobile robots that must avoid static and dynamic obstacles. Zhang et al. [148] used this kind of technique to find the shortest path in maze scenarios. This approach has also been used together with genetic algorithms [149,150].

Other works in the literature combine fuzzy logic and neural networks [151–154]. There have been different approaches to this. For instance, Mohanty and Parhi [155] use many of these systems for autonomous navigation. Further insights and a more extensive surveys on related hybrid methods are provided in Mac et al. [8]. Furthermore, the use of Reinforcement Learning (RL) has also been studied to control the motion of a robot [156,157]. Faust et al. [158] combined RL with the Probabilistic Roadmap Method (PRM), which is one of the algorithms detailed next. For more information about planning algorithms based on RL, refer to the work of Sun et al. [21].

5. C-Space-Search-Based Path Planning Algorithms

Algorithms in this category consider the working space of the path planner as the space of all states or configurations reachable by the robot. For this reason, most of the works in this category refer to this working space as the *C-Space*. The main idea behind these algorithms is to use a discrete set of samples that are part of this *C-Space*. In other words, the *C-Space* is discretized. This set of samples includes the initial and the goal states, or at least samples relatively close to them. In this way, these algorithms execute a search operation, visiting samples from this set. At a certain point, the algorithm will find and return a certain subset of samples connecting the initial and goal states representing the resulting path. In other words, the *waypoints* forming the paths correspond, each of them, to a sample from the C-Space. This implies that the generated path heavily depends on how these samples are scattered, how they are connected and how are they visited. In fact, due to this dependency, in some approaches post-processing is done to smooth the shape of the resulting path.

The *C-Space Search* category is subdivided into two groups of algorithms according to how they discretize the *C-Space*. *Graph Search* algorithms do this using a pre-existing graph (such as one of those depicted in Figure 2). Each of the nodes from this graph represents a *C-Space* sample and is connected to other nearby nodes, i.e., its neighbors. With regards to *Sampling-Based* algorithms, they focus on the creation and/or modification of samples within the *C-Space* in an iterative way. They can keep working, even after finding a feasible path, to find better ones.

5.1. Graph Search

As mentioned, the *C-Space* can be discretized in the form of a graph. *Graph Search* algorithms fully or partially visit this graph until they find a path connecting the initial and goal states. The first algorithms made in this category return paths of which the *waypoints* are placed on top of neighboring samples. In other words, the connections between consecutive *waypoints* of the path are coincident with the graph edges. Figure 8a depicts a schematic showing how in the first case the shape of the path is determined by these edges. This review, therefore, categorizes the *Graph Search* algorithms generating this kind of path as *Edge-restricted*. These paths consequently depend on how the graph is structured. As seen in Section 2.2.1 there are various graph structures in the form of *Cell Decomposition* and *Roadmaps*. Another kind of *Graph Search* algorithm, *Any-angle*, were created to solve this issue with the restriction to the graph edges. The reason this name is used is that paths produced by *Edge-restricted* planners only use certain values of orientation. For example, in an eight-neighborhood regular grid, as in the case shown in Figure 8a, *Edge-restricted* paths can only have orientations with 0, ±45, ±90, ±135 and 180 degrees. *Any-angle* algorithms produce paths that are not restricted to these orientations as their *waypoints* do not necessarily have to be placed in neighboring nodes. Figure 8b depicts an example of this.

神经网络，Zou 等人。 [146]对这种算法在2000年代初的应用进行了简要的调查。 Engedy 和 Horváth [147] 提出了一种使用神经网络的路径规划器，用于必须避开静态和动态障碍的移动机器人。张等人。 [148]使用这种技术来寻找迷宫场景中的最短路径。这种方法也已与遗传算法一起使用[149,150]。

文献中的其他作品结合了模糊逻辑和神经网络[151-154]。对此有不同的方法。例如，Mohanty 和 Parhi [155] 使用许多此类系统进行自主导航。 Mac 等人提供了对相关混合方法的进一步见解和更广泛的调查。 [8]。此外，还研究了使用强化学习（RL）来控制机器人的运动[156,157]。浮士德等人。 [158]将强化学习与概率路线图方法（PRM）结合起来，这是接下来详细介绍的算法之一。有关基于 RL 的规划算法的更多信息，请参阅 Sun 等人的工作。 [21]。

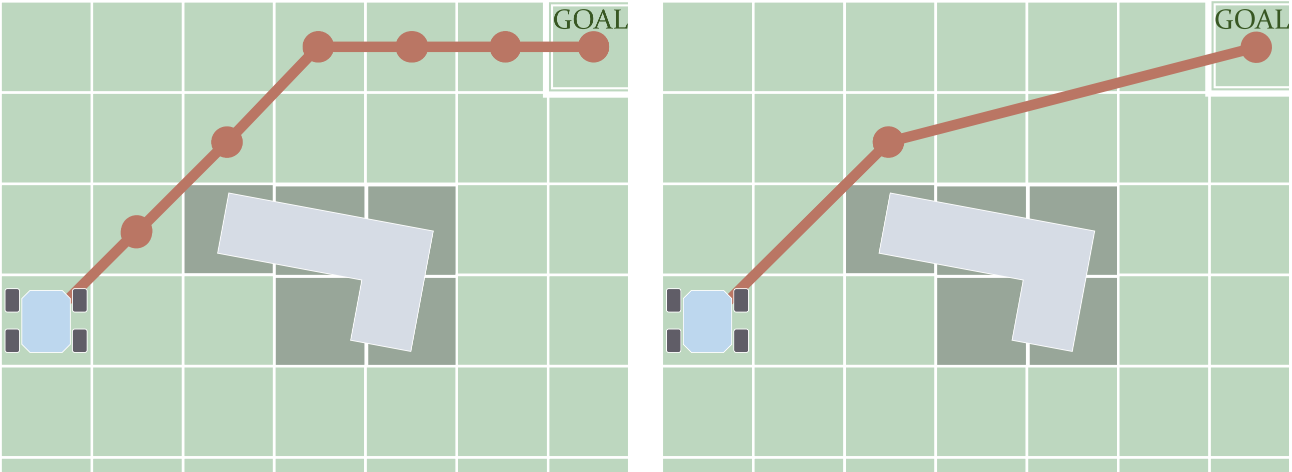
5. 基于C空间搜索的路径规划算法

此类算法将路径规划器的工作空间视为机器人可到达的所有状态或配置的空间。因此，该类别的大多数作品都将这个工作空间称为*C-Space*。这些算法背后的主要思想是使用属于此 *C-Space* 的一组离散样本。换句话说，*C-Space* 被离散化。这组样本包括初始状态和目标状态，或者至少是与它们相对接近的样本。通过这种方式，这些算法执行搜索操作，访问该集合中的样本。在某一点，算法将找到并返回连接表示结果路径的初始状态和目标状态的特定样本子集。换句话说，形成路径的 *waypoints* 对应于 C 空间中的每个样本。这意味着生成的路径在很大程度上取决于这些样本的分散方式、连接方式以及访问方式。事实上，由于这种依赖性，在某些方法中进行后处理以平滑结果路径的形状。

根据离散化 *C-Space* 的方式，*C-Space Search* 类别被细分为两组算法。 *Graph Search* 算法使用预先存在的图（如图 2 所示的图之一）来完成此操作。该图中的每个节点代表一个 *C-Space* 样本，并连接到其他附近的节点，即其邻居。对于 *Sampling-Based* 算法，它们专注于以迭代方式创建和/或修改 *C-Space* 中的样本。即使找到了一条可行的路径，他们也可以继续努力，找到更好的路径。

5.1. Graph Search

如前所述，*C-Space* 可以以图的形式离散化。 *Graph Search* 算法完全或部分访问该图，直到找到连接初始状态和目标状态的路径。此类别中的第一个算法返回路径，其中 *waypoints* 放置在相邻样本的顶部。换句话说，路径的连续*waypoints*之间的连接与图的边重合。图 8a 描绘了示意图，显示在第一种情况下如何由这些边缘确定路径的形状。因此，本评论将生成此类路径的 *Graph Search* 算法分类为 *Edge-restricted*。因此，这些路径取决于图的结构。如第 2.2.1 节所示，存在 *Cell Decomposition* 和 *Roadmaps* 形式的各种图结构。另一种 *Graph Search* 算法 *Any-angle* 是为了解决这个限制图边的问题而创建的。使用此名称的原因是 *Edge-restricted* 规划器生成的路径仅使用某些方向值。例如，在八邻域规则网格中，如图 8a 所示的情况，*Edge-restricted* 路径只能具有 0、±45、±90、±135 和 180 度的方向。 *Any-angle* 算法生成的路径不限于这些方向，因为它们的 *waypoints* 不一定必须放置在相邻节点中。图 8b 描述了一个例子。

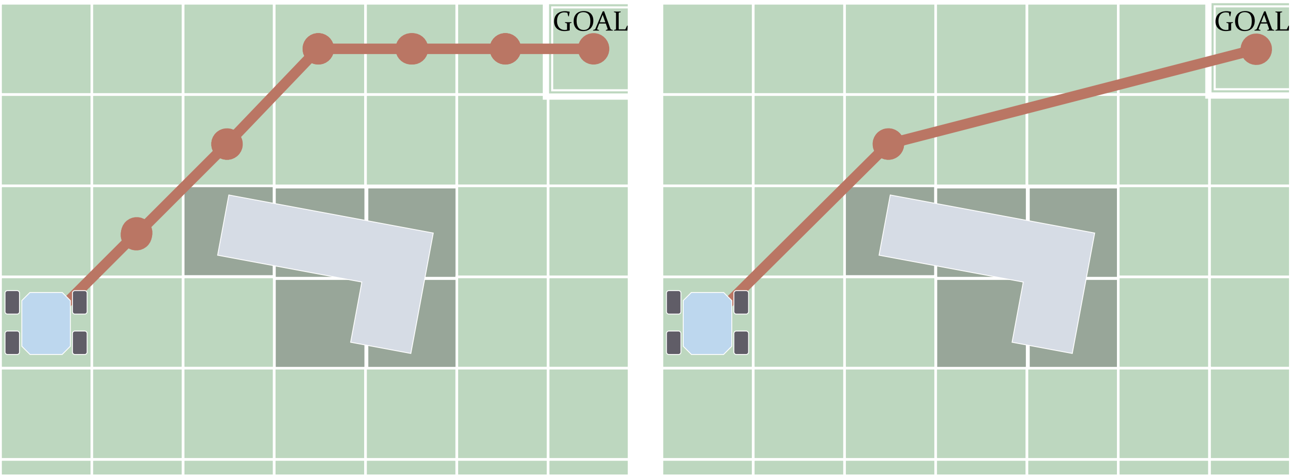


(**a**) (**b**)

Figure 8. Main difference between the paths (in red) produced by *Edge-restricted* (**a**) and *Any-angle* algorithms (**b**): in the first case, *waypoints* can be placed only on consecutive (neighboring) nodes.

With regards to *Edge-restricted* algorithms, Figure 9 shows a schematic with the most representative versions of them that can be found in the literature. The most known and basic *Edge-restricted* path planner is the Dijkstra algorithm [159]. As an initial step, this algorithm takes one node, either the start or the goal. Thereafter, it proceeds to propagate information to its neighbors. It could be either the value of the cost required to arrive from the start or the one that remains to arrive at the goal. Iteratively, the algorithms visit the neighbors of already-visited nodes. The information about the amount of cost keeps propagating, and the algorithms assign to each visited node a parent node. If the environment allows it, i.e., no obstacles are isolating either the goal or the start, the algorithm ends up visiting both of them. At this point, the path is retrieved by backtracking the parent nodes. In other words, the path starts from the last visited node and goes back through the parent nodes. Years later, Hart et al. [160] implemented a heuristic version, A*, to speed up computation. Later on, further improvements were introduced. D*, also named Dynamic A*, was introduced by [161] as an incremental version of A*. The fact that it is incremental means that this algorithm recycles previous computations whenever there are changes in the cost assigned to the grid nodes. This prevents the algorithm from executing a whole new computation from scratch. This reduction in the computation allows for rapid replanning in cases where the robot encounters novel obstacles on its way, for example. An improved version called Focussed D* managed to further reduce the computation time of D* [162]. Koenig and Likhachev [163] propose the use of Lifelong Planning A* (LPA*) as another direct incremental extension of A*. This was taken into account as a reference to produce a much simpler version of D* called D*-Lite [164,165]. Colas et al. [166] employed this algorithm on mobile robots in search-and-rescue applications. Since A* and D* algorithms, including their versions, make use of heuristic functions, the resulting paths can be sub-optimal. Likhachev et al. [167] propose anytime versions of these algorithms, which use a configurable fixed time. The best path found, at a given time, is generated by these anytime versions. Dolgov et al. [168] propose a version of A*, *Hybrid* A*, that prioritizes the feasibility of the resulting paths in exchange for the loss of optimality and completeness by rearranging the nodes after a path is found; in a way, the path is kinematically feasible.

With regards to the *Any-angle* algorithms, one of the first ones was Field-D* [169]. This is a well-known algorithm mainly due to its use on Mars NASA rovers since *Spirit* and *Opportunity* [170]. Similarly to D* and D*-lite, it is an incremental algorithm, so it recycles previous computations in subsequent executions. Although Field-D* arose as an outstanding method to overcome the problem of paths restricted to edges, there was still a margin to improve the results and find even more optimal paths. Years later, more *Any-angle* algorithms were created having A* as a basis, focused on the problem of finding



(**a**) (**b**)

图 8. *Edge-restricted* (a) 和 *Any-angle* 算法生成的路径（红色）之间的主要区别 (b)：在第一种情况下，*waypoints* 只能放置在连续（相邻）节点上。

关于 *Edge-restricted* 算法，图 9 显示了示意图，其中包含可在文献中找到的最具代表性的版本。最著名和基本的 *Edge-restricted* 路径规划器是 Dijkstra 算法 [159]。作为第一步，该算法采用一个节点，即起点或目标。此后，它继续向其邻居传播信息。它可以是从一开始所需的成本值，也可以是达到目标所需的成本值。算法迭代地访问已访问节点的邻居。有关成本量的信息不断传播，算法为每个访问的节点分配一个父节点。如果环境允许，即没有障碍物隔离目标或起点，算法最终会访问这两个目标。此时，通过回溯父节点来检索路径。换句话说，该路径从最后访问的节点开始并通过父节点返回。几年后，哈特等人。 [160]实现了启发式版本 A*，以加速计算。后来又进行了进一步的改进。 D*，也称为 Dynamic A*，由[161]作为 A* 的增量版本引入。事实上，它是增量的，这意味着只要分配给网格节点的成本发生变化，该算法就会回收以前的计算。这可以防止算法从头开始执行全新的计算。例如，计算量的减少允许在机器人在途中遇到新障碍物的情况下快速重新规划。一个名为 Focussed D* 的改进版本设法进一步减少了 D* 的计算时间 [162]。 Koenig 和 Likhachev [163] 提议使用终身规划 A* (LPA*) 作为 A* 的另一种直接增量扩展。这被考虑作为参考来生成一个更简单的 D* 版本，称为 D*-Lite [164,165]。可拉斯等人。 [166]在搜索和救援应用中的移动机器人上采用了这种算法。由于 A* 和 D* 算法（包括它们的版本）使用启发式函数，因此生成的路径可能不是最优的。利哈乔夫等人。 [167]提出了这些算法的任何时间版本，它使用可配置的固定时间。在给定时间找到的最佳路径是由这些随时版本生成的。多尔戈夫等人。 [168] 提出了 A* 的一个版本，*Hybrid* A*，它优先考虑结果路径的可行性，以换取在找到路径后重新排列节点以损失最优性和完整性；在某种程度上，该路径在运动学上是可行的。

关于 *Any-angle* 算法，最早的算法之一是 Field-D* [169]。这是一种众所周知的算法，主要是因为它自 *Spirit* 和 *Opportunity* [170] 以来就在 NASA 火星漫游车上使用。与 D* 和 D*-lite 类似，它是一种增量算法，因此它会在后续执行中回收以前的计算。尽管 Field-D* 是一种克服边缘受限路径问题的出色方法，但在改进结果和找到更优化路径方面仍有余地。几年后，更多以 A* 为基础的 *Any-angle* 算法被创建，重点关注寻找问题

the shortest paths while avoiding obstacles. Nash et al. [171] created Theta* for this purpose. They presented it in two versions: one computationally cheaper, *Base*-Theta*, and another more expensive but with results closer to the globally optimal shortest path, named *Angle-Propagation* Theta*. The main premise behind Theta* was the consideration of heading changes on obstacle corners, reducing these changes along the path in contrast with Field-D* [172]. An incremental version of the *Base*-Theta* algorithm was later introduced with the name of Incremental Phi* [173]. A faster version of Theta*, *Lazy*-Theta*, was introduced by Nash et al. [174]. Theta* was also improved to work better on non-uniform cost maps, where the cost increases the closer it gets to obstacles [175]. The Accelerated A* algorithm was introduced by Šišlák et al. [176]. It finds shorter paths than Theta* but at a slower rate (although still faster than A*). Yap et al. [177] also introduced another algorithm named Block A* and compared its performance with A* and Theta*. Another comparative study including Accelerated A*, Block A*, Field-D* and Theta* was provided by Nash and Koenig [11], also including variants of the latter algorithm. The same authors later compared this algorithm with the use of visibility graphs [178]. Muñoz and R-Moreno [179] proposed the use of S-Theta* to produce paths that smooth the heading changes. 3DANA is used to produce paths on elevation maps [180,181]. Other algorithms that have shown improved results compared to Theta* and many other *Any-angle* algorithms include *Any-angle* Subgoal Graphs [182,183] and Anya [184,185].

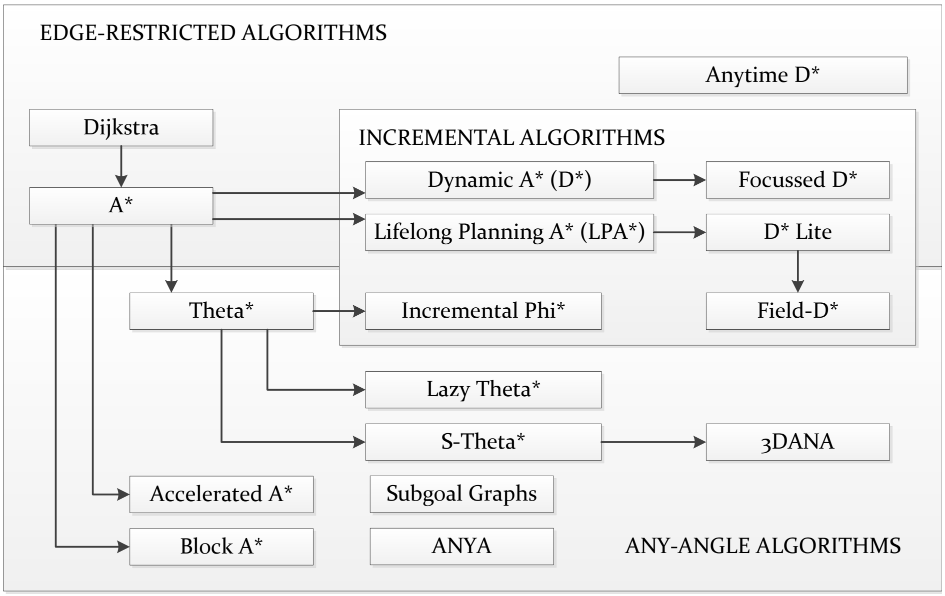


Figure 9. Overview of many different approaches to *Graph Search* path planning. The arrows indicate how most of them rest on older approaches yet introduce significant improvements.

5.2. Sampling-Based

Sampling-Based path planning algorithms create samples of the C-Space one after another, following different policies [12,186]. Later on, they retrieve the path from the created samples after meeting a certain condition or set of conditions, such as reaching a time limit. This kind of algorithm is asymptotically optimal. It means they can create more and more samples, attempting to find a better solution as time goes on. In general, these algorithms are usually used for searches in high-dimensional spaces. However, the number of samples may be relatively large in order to get close to the global optimal solution [14], demanding the use of large memory resources to store all the samples.

If only two points are considered (the starting position or state and goal), the algorithm is a single-query algorithm, whereas if more points are selected for the same environment then the algorithm is categorized as multiple-query. With regards to the single-query, one of the most famous is the Rapidly Random Tree (RRT) algorithm, which is also a special case of

避开障碍物的同时寻找最短路径。纳什等人。 [171] 为此目的创建了 Theta*。他们提供了两个版本：一种是计算成本较低的 *Base*-Theta*，另一种是成本较高但结果更接近全局最优最短路径的版本，称为 *Angle- Propagation* Theta*。 Theta* 背后的主要前提是考虑障碍角上的航向变化，与 Field-D* [172] 相比，减少了沿路径的这些变化。后来引入了 *Base*-Theta* 算法的增量版本，名称为增量 Phi* [173]。 Nash 等人引入了 Theta* 的更快版本，*Lazy*-Theta*。 [174]。 Theta* 也经过改进，可以在非均匀成本地图上更好地工作，其中距离障碍物越近，成本就会增加 [175]。 Accelerated A* 算法由 i lák 等人提出。 [176]。它找到比 Theta* 更短的路径，但速度较慢（尽管仍然比 A* 快）。叶等人。 [177]还介绍了另一种名为Block A*的算法，并将其性能与A*和Theta*进行了比较。 Nash 和 Koenig [11] 提供了另一项比较研究，包括 Accelerated A*、Block A*、Field-D* 和 Theta*，还包括后一种算法的变体。同一作者后来将该算法与可见性图的使用进行了比较[178]。 Muñoz 和 R-Moreno [179] 提议使用 S-Theta* 来生成平滑航向变化的路径。 3DANA 用于在高程地图上生成路径 [180,181]。与 Theta* 和许多其他 *Any-angle* 算法相比，其他算法显示出改进的结果，包括 *Any-angle* Subgoal Graphs [182,183] 和 Anya [184,185]。

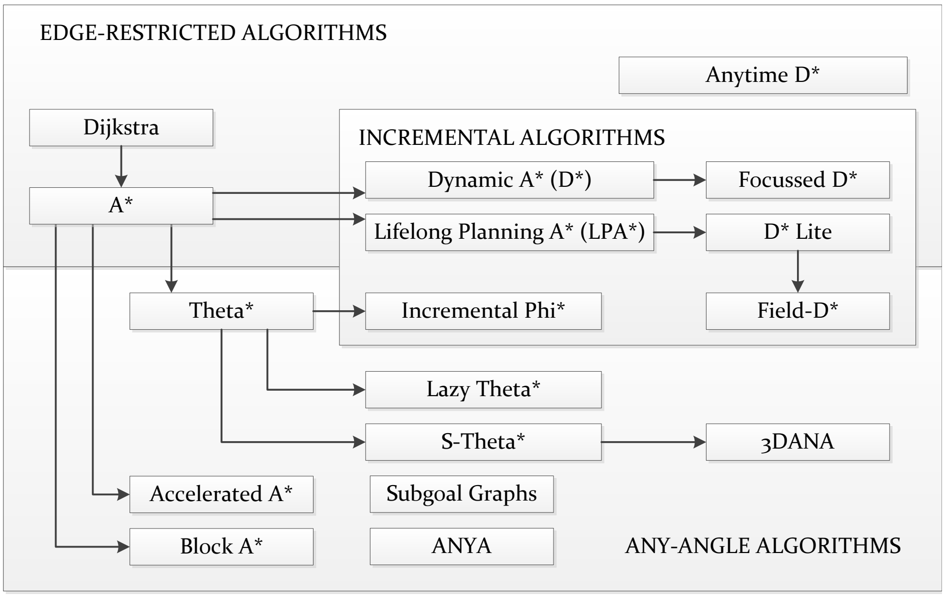


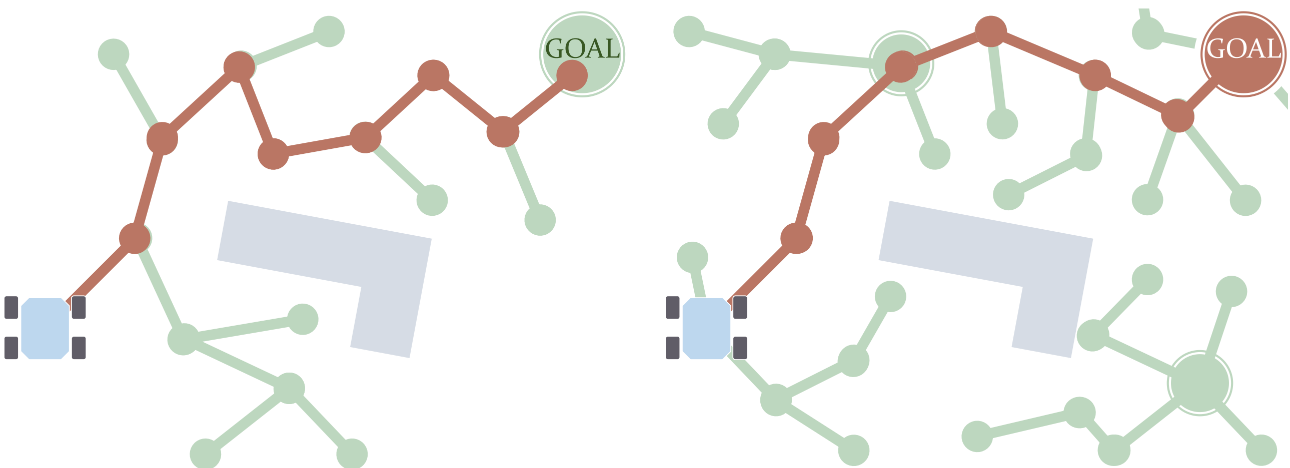
图 9. *Graph Search* 路径规划的多种不同方法的概述。箭头表明它们中的大多数如何依赖于旧方法，但又引入了重大改进。

5.2. *Sampling-Based*

Sampling-Based 路径规划算法遵循不同的策略，一个接一个地创建 C 空间的样本 [12, 186]。随后，他们在满足特定条件或一组条件（例如达到时间限制）后从创建的样本中检索路径。这种算法是渐近最优的。这意味着他们可以创建越来越多的样本，随着时间的推移尝试找到更好的解决方案。一般来说，这些算法通常用于高维空间中的搜索。然而，为了接近全局最优解，样本数量可能会比较大[14]，需要使用大量的内存资源来存储所有样本。

如果仅考虑两个点（起始位置或状态和目标），则该算法是单查询算法，而如果为同一环境选择更多点，则该算法被归类为多查询算法。对于单查询，最著名的是快速随机树（RRT）算法，它也是

the Rapidly Deterministic Tree (RDT) [187]. This algorithm emulates a tree growing in the sense that from a starting point the samples are dynamically created as if they were branches. Figure 10a depicts an scheme summarizing this process. When one of the samples is closer to the goal than a certain distance, then the path can be retrieved by tracking backwards until reaching the origin point. As mentioned, more iterations can still be executed to find better paths. Further modifications of RRT can be found in the literature. A bi-directional version was introduced by Kuffner and LaValle [188] with the name of RRT-Connect. Later on, Yershova et al. [189] presented an improved version of RRT called *Dynamic Domain* RRT, which was aware of the obstacles existing in the environment during the expansion of the tree. Arslan and Tsiotras [190] took ideas from the *Graph Search* algorithm LPA to make RRT#, an improved version of RRT with a faster convergence rate. Karaman and Frazzoli [186] introduced a heuristic version of RRT named Heuristic RRT (RRT*) to speed up computation while still being asymptotically optimal. For an extensive review of RRT* variants, refer to the work of Noreen et al. [14]. An improved version called Informed RRT* improved upon the performance of RRT* by delimiting an ellipse enclosing the start and goal positions when a feasible path is found [191]. The next iterations to improve this path are done within this ellipse, instead of making the algorithm explore other options that probably will not influence the result. Gammell et al. [192] presented another improvement to this, called Batch Informed Trees (BIT*), together with a comparison demonstrating its better performance than RRT*, Informed RRT* and FMT*. BIT* also takes some steps from the *Graph Search* algorithm LPA*. Regionally Accelerated Batch Informed Trees (AIT*) improved upon BIT*, especially for cases where narrow corridors exist [193]. The Adaptively Informed Trees (AIT*) and Advanced Batch Informed Trees (ABIT*) algorithms improved upon the BIT* algorithm and were integrated into an experimental NASA rover [194,195].

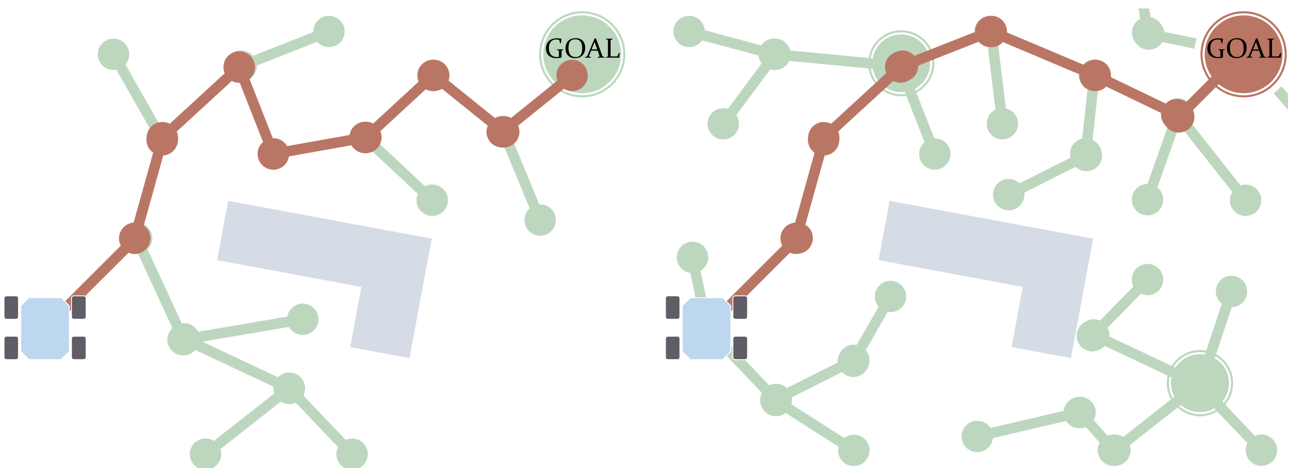


(**a**) (**b**)

Figure 10. Example cases of single-query (**a**) and multiple-query (**b**) *Sampling-Based* algorithms. Samples are created in an iterative way until the destination is reached.

With regards to multiple-query *Sampling-Based* algorithms, the most famous is the Probabilistic Roadmap Method (PRM) [196]. This algorithm starts with a series of samples that are already scattered over the *C-Space*. From here, new samples are created, creating a new tree from each of these initial samples. Figure 10b illustrates the concept behind this process. Thereafter, a *Graph Search* method such as A* is used to retrieve the path using the graph created by PRM. Karaman and Frazzoli [186] introduced a heuristic version of PRM. The improvement proposed by Park et al. [197] uses a hierarchical structure that reduces the number of samples. PRM has been tested in robots within simulated indoor scenarios in the work of Alenezi et al. [1]. Ichter et al. [198] presented *Critical* PRM, an algorithm that combines PRM with *Reinforcement Learning* to determine critical locations such as narrow corridors.

快速确定树（RDT）[187]。该算法模拟树的生长，从起点开始动态创建样本，就好像它们是分支一样。图 10a 描绘了总结此过程的方案。当其中一个样本距离目标较近某一距离时，则可以通过向后跟踪直到到达原点来检索路径。如前所述，仍然可以执行更多迭代来找到更好的路径。 RRT 的进一步修改可以在文献中找到。 Kuffner 和 LaValle [188] 引入了双向版本，名为 RRT-Connect。后来，Yershova 等人。 [189]提出了 RRT 的改进版本，称为 *Dynamic Domain* RRT，它在树的扩展过程中意识到环境中存在的障碍。 Arslan 和 Tsiotras [190] 从 *Graph Search* 算法 LPA 中汲取灵感，提出了 RRT#，这是 RRT 的改进版本，具有更快的收敛速度。 Karaman 和 Frazzoli [186] 引入了 RRT 的启发式版本，名为 Heuristic RRT (RRT*)，以加速计算，同时仍保持渐近最优。有关 RRT* 变体的广泛回顾，请参阅 Noreen 等人的工作。 [14]。称为 Informed RRT* 的改进版本通过在找到可行路径时划定一个包围起始位置和目标位置的椭圆来改进 RRT* 的性能 [191]。改进这条路径的下一次迭代是在这个椭圆内完成的，而不是让算法探索可能不会影响结果的其他选项。加梅尔等人。 [192] 对此提出了另一种改进，称为批量知情树 (BIT*)，并通过比较证明其比 RRT*、知情 RRT* 和 FMT* 更好的性能。 BIT* 还采用了 *Graph Search* 算法 LPA* 的一些步骤。区域加速批量知情树 (AIT*) 改进了 BIT*，特别是对于存在狭窄走廊的情况 [193]。自适应信息树 (AIT*) 和高级批量信息树 (ABIT*) 算法在 BIT* 算法的基础上进行了改进，并集成到实验性 NASA 漫游车中 [194,195]。



(a) **(b)**

图 10. 单查询 (a) 和多查询 (b) *Sampling-Based* 算法的示例。样本以迭代方式创建，直到到达目的地。

关于多查询*Sampling-Based*算法，最著名的是概率路线图方法（PRM）[196]。该算法从一系列已经分散在 *C-Space* 上的样本开始。从这里开始，创建新样本，从每个初始样本创建一棵新树。图 10b 说明了该过程背后的概念。此后，使用诸如 A* 之类的 *Graph Search* 方法来使用 PRM 创建的图来检索路径。 Karaman 和 Frazzoli [186] 引入了 PRM 的启发式版本。 Park 等人提出的改进。 [197]使用层次结构来减少样本数量。 Alenezi 等人的工作中，PRM 已在模拟室内场景中的机器人中进行了测试。 [1]。伊希特等人。 [198]提出了*Critical* PRM，一种将PRM与*Reinforcement Learning*相结合的算法，以确定狭窄走廊等关键位置。

Another *Sampling-Based* algorithm, named the Fast Marching Tree (FMT*) algorithm, was created to reduce the convergence rate of both RRT and PRM. It takes features from not only the two of them but also an *Optimal Control* algorithm called FMM, which is detailed below. The main objective of FMT* is to find paths, avoiding obstacles, in problems involving a high number of degrees of freedom. One example of this is the motion planning of an articulated vehicle presented by Reid et al. [50]. Ichter et al. [199] propose the use of Group Marching Tree (GMT*), a similar algorithm to FMT* but which focuses on speeding up computation via parallelization using GPUs. Finally, it is worth mentioning there are path planning algorithms that combine the *Dynamic Sampling* approach with Model Predictive Control (MPC) techniques to account for kinodynamic constraints [57,200,201].

6. Optimal-Control-Based Path Planning Algorithms

The baseline of *Control Approach* based algorithms is the creation of a control function that takes the robot from an initial state in the C-Space to the destination. As the name suggests, here the path planning problem is addressed using an optimal control approach [202]. The main difference with *Soft Computation* methods is that there are no configurable parameters; the problem here must be fully enclosed. Here, there are two different subcategories. In the first of them, *PDE Solving*, algorithms solve a Partial Derivative Equation (PDE) on a grid, based on the Dynamic Programming Principle (DPP) [203]. The second subcategory, *Numerical Optimization*, encompasses algorithms that in general optimize a pre-existing path given the kino-dynamic restrictions of the robot to make it feasible.

6.1. PDE-Solving-Based

The optimal control approach is based here on the Dynamic Programming Principle through the resolution of the Hamilton–Jacobi–Bellman equation [204] using a grid. Since it is a partial derivative equation (PDE), this sub-category is named *PDE Solving*. It can be seen as finding the numerical solution to the problem of calculating the propagation of a wave over a grid. A value of the wave arrival time is assigned to each of the grid nodes. The way the wave propagates will depend on how the HJB equation is formulated, including the cost function. The main drawback of this kind of algorithm is that it generally cannot deal with constraints in the form of discontinuities.

A particular case of the HJB equation is the *Eikonal* equation. This is not only static but also considers the cost function and only returns a scalar value according to the position on the map. This means the wave propagates on a node with a rate that depends only on the assigned scalar value. In this way, the characteristic directions are coincident with the gradient of the *Total Cost* function, and hence the path can be retrieved simply using a *Gradient Descent* method. A whole family of methods have been proposed over the years to compute the solution to this problem-formulation with low computational requirements, and hence they are named *Fast* methods [18]. One of the most well-known is the Fast Marching Method (FMM), introduced by Sethian [205]. This algorithm follows the same strategy as Dijkstra to visit the nodes of a grid. Unlike Dijkstra, FMM assigns the value of an amount of cost to each node by solving the *Eikonal* algorithm. The resulting path is smooth, continuous and optimal. Chiang et al. [206] compared this algorithm with A* and demonstrated how, as the path is not restricted by the grid, FMM gets shorter paths. There are many existing works using FMM for path planning [207–210]. Some of its variants, most of them reducing the computational power requirements of FMM, are introduced in the review by Gómez et al. [18], along with other *Eikonal* solvers. They are *Binary* FMM, *Fibonacci* FMM, *Simplified* FMM and *Untidy* FMM. The cost values used by the *Eikonal* determine the rate of propagation of the computed wave. Petres et al. [24] demonstrated how the gradient of these cost values affects the curvature radius of the resulting path. Their work also introduced the idea of Heuristic Fast Marching (FM*) and also compares this with other *Graph Search* algorithms. For smooth obstacle avoidance, Fast Marching Square (FMS) [74] computes FMM twice, being the first to create a repulsive field surrounding obstacles. Figure 11a depicts how the path smoothly gets further from the

另一种 *Sampling-Based* 算法称为快速行进树 (FMT*) 算法，旨在降低 RRT 和 PRM 的收敛速度。它不仅采用了两者的特征，还采用了一种称为 FMM 的 *Optimal Control* 算法，下面将详细介绍。 FMT* 的主要目标是在涉及高自由度的问题中寻找路径、避开障碍。其中一个例子是 Reid 等人提出的铰接式车辆的运动规划。 [50]。伊希特等人。 [199] 提出使用 Group Marching Tree (GMT*)，这是一种与 FMT* 类似的算法，但其重点是通过使用 GPU 并行化来加速计算。最后，值得一提的是，有一些路径规划算法将 *Dynamic Sampling* 方法与模型预测控制 (MPC) 技术相结合，以考虑运动动力学约束 [57,200,201]。

6. 基于最优控制的路径规划算法

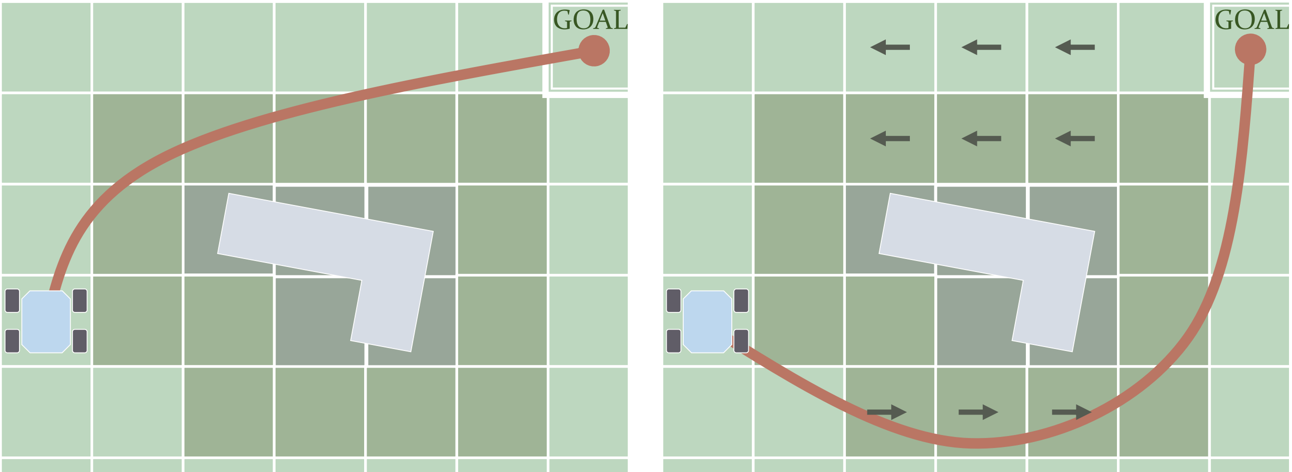
基于 *Control Approach* 的算法的基线是创建一个控制函数，该函数使机器人从 C 空间中的初始状态到达目的地。顾名思义，这里的路径规划问题是使用最优控制方法来解决的[202]。与 *Soft Computation* 方法的主要区别在于没有可配置的参数；这里的问题必须被完全封闭。这里有两个不同的子类别。在第一个 *PDE Solving* 中，算法基于动态规划原理 (DPP) [203] 在网格上求解偏导方程 (PDE)。第二个子类别 *Numerical Optimization* 包含的算法通常会在给定机器人的运动动力学限制的情况下优化预先存在的路径，使其可行。

6.1. PDE-Solving-Based

此处的最优控制方法基于动态规划原理，通过使用网格求解 Hamilton-Jacobi-Bellman 方程 [204]。由于它是偏导方程 (PDE)，因此该子类别被命名为 *PDE Solving*。它可以被视为寻找计算波在网格上传播的问题的数值解。波到达时间的值被分配给每个网格节点。波的传播方式取决于 HJB 方程的公式，包括成本函数。这种算法的主要缺点是它通常无法处理不连续形式的约束。

HJB 方程的一个特殊情况是 *Eikonal* 方程。这不仅是静态的，而且还考虑了成本函数，仅根据地图上的位置返回标量值。这意味着波在节点上传播的速率仅取决于指定的标量值。这样，特征方向与*Total Cost*函数的梯度一致，因此可以简单地使用*Gradient Descent*方法来检索路径。多年来，人们提出了一整套方法来计算这个问题的解决方案，计算要求较低，因此它们被命名为 *Fast* 方法 [18]。其中最著名的方法之一是由 Sethian [205] 提出的快速行进方法 (FMM)。该算法遵循与 Dijkstra 相同的策略来访问网格的节点。与Dijkstra不同的是，FMM通过求解*Eikonal*算法为每个节点分配一定量的成本值。由此产生的路径是平滑、连续且最佳的。蒋等人。 [206]将该算法与A*进行了比较，并证明了由于路径不受网格限制，FMM如何获得更短的路径。有许多现有的工作使用 FMM 进行路径规划 [207-210]。Gómez 等人在评论中介绍了它的一些变体，其中大多数变体降低了 FMM 的计算能力要求。[18]，以及其他 *Eikonal* 求解器。它们是*Binary* FMM、*Fibonacci* FMM、*Simplified* FMM 和*Untidy* FMM。 *Eikonal* 使用的成本值决定了计算波的传播速率。佩特雷斯等人。 [24]演示了这些成本值的梯度如何影响结果路径的曲率半径。他们的工作还介绍了启发式快速行进 (FM*) 的思想，并将其与其他 *Graph Search* 算法进行了比较。为了平滑避障，Fast Marching Square (FMS) [74] 计算 FMM 两次，这是第一个在障碍物周围创建排斥场的方法。图 11a 描述了路径如何平滑地远离

obstacle thanks to the higher values of cost near it (darker colors). Liu and Bucknall [211] used FMS, along with a modification of the cost around the initial position, to consider the initial orientation of the vehicle. Researchers have also aimed to propose incremental versions of FMM, such as E* [212–214].



(**a**) (**b**)

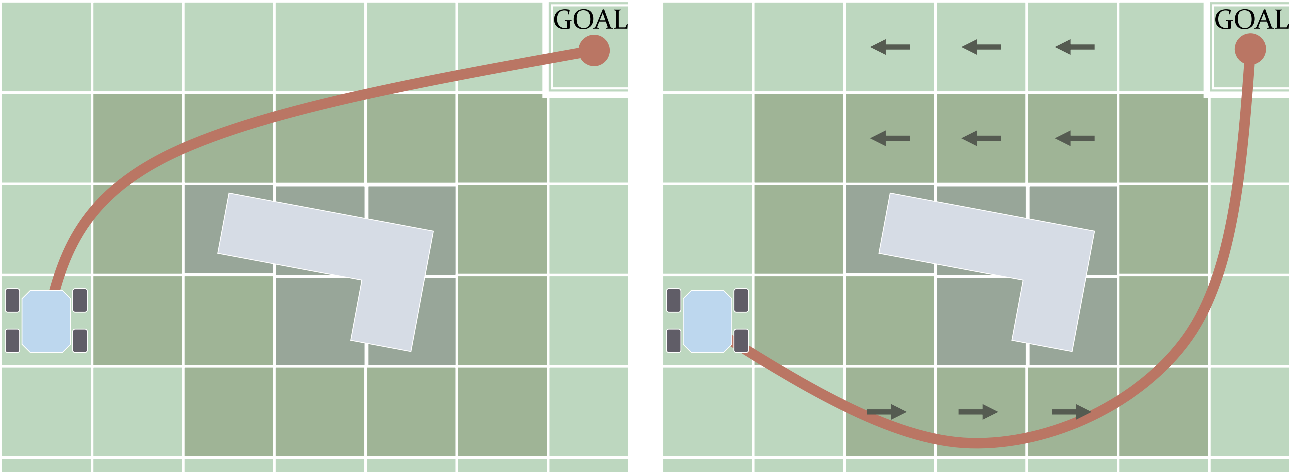
Figure 11. *PDE-Solving*-based algorithms can calculate a continuous and smooth path in non-uniform cost maps. The cost assigned to each cell can be a scalar value or isotropic (**a**) or in the form of a vector, i.e., anisotropic (**b**).

To work with more general expressions of the Hamilton–Jacobi-0Bellman (HJB) equation, other kinds of methods must be used. FMM produces sub-optimal results if used with direction-dependent (anisotropic) costs [215]. This kind of cost implies that the wave propagates differently depending on its direction relative to how a vectorial cost is assigned to the node. There are particular situations in which FMM produces accurate results under a certain level of anisotropy, such as having a cost function formulated in such a way thatit varies mostly in the directions parallel to the reference axes [24,216]. This is the case depicted in Figure 11b. Sethian and Vladimirsky [215] proposed the use of an algorithm called the Ordered Upwind Method (OUM) to deal with the static HJB equation, the convergence rate of which was demonstrated by Shum et al. [217]. Its main drawback is the increase in computational cost it entails, proportional to the anisotropy existing in the scenario. Shum et al. [31] used HJB for anisotropic path planning, considering energy minimization and stability, and considering the direction and magnitude of slopes. The Fast Sweeping Method (FSM) has also been demonstrated to work with general static HJB equations [218]. It works by visiting all nodes on a grid, following certain directions repeatedly, which means that it demands a high number of iterations. Takei and Tsai [37] used FSM to formulate the HJB equation to comply with turning radius constraints. For the *Eikonal* case, Bak et al. [219] introduced an improvement to FSM to speed up its computation when the cost varies too much, and Detrixhe et al. [220] introduced a parallel version. Jeong and Whitaker [221] proposed an algorithm named the Fast Iterative Method (FIM) to solve the *Eikonal* equation on parallel architectures as well.

6.2. Global Optimization

This subcategory contains path planning algorithms that optimize an existing preliminary feasible path. Unlike *Local Optimization* methods, introduced in Section 3.1, *Global Optimization* methods make the resulting path globally optimal in exchange for investing more of a computational load. The approach presented by Ratliff et al. [222], for instance, uses *Sampling-Based* methods such as RRT or PRM as a first step. The second step consists of using gradient optimization techniques to approximate the optimal solution from this feasible path. Van Den Berg et al. [223] also started with a trajectory computed using RRT, to later apply jt to an optimization process based on Differential Dynamic Program-

由于附近的成本值较高（颜色较深），因此出现了障碍。 Liu 和 Bucknall [211] 使用 FMS 以及围绕初始位置的成本修改来考虑车辆的初始方向。研究人员还致力于提出 FMM 的增量版本，例如 E* [212-214]。



(**a**) (**b**)

图 11. 基于 *PDE-Solving* 的算法可以计算非均匀成本图中的连续且平滑的路径。分配给每个单元的成本可以是标量值或各向同性 (a) 或矢量形式，即各向异性 (b)。

为了使用 Hamilton-Jacobi-0Bellman (HJB) 方程的更一般的表达式，必须使用其他类型的方法。如果与方向相关（各向异性）成本一起使用，FMM 会产生次优结果 [215]。这种成本意味着波的传播方式不同，具体取决于其相对于矢量成本如何分配给节点的方向。在某些特殊情况下，FMM 在一定水平的各向异性下产生准确的结果，例如以这种方式制定的成本函数主要在平行于参考轴的方向上变化[24,216]。图 11b 中描述的就是这种情况。 Sethian 和 Vladimirsky [215] 提出使用一种称为有序迎风法 (OUM) 的算法来处理静态 HJB 方程，Shum 等人证明了其收敛速度。 [217]。它的主要缺点是计算成本的增加，与场景中存在的各向异性成正比。舒姆等人。 [31]使用HJB进行各向异性路径规划，考虑能量最小化和稳定性，并考虑斜坡的方向和大小。快速扫描方法 (FSM) 也已被证明适用于一般静态 HJB 方程 [218]。它的工作原理是重复遵循某些方向，访问网格上的所有节点，这意味着它需要大量迭代。 Takei 和 Tsai [37] 使用 FSM 制定 HJB 方程以满足转弯半径约束。对于 *Eikonal* 情况，Bak 等人。 [219]引入了对 FSM 的改进，以在成本变化太大时加快计算速度，Detrixhe 等人。 [220]引入了并行版本。 Jeong 和 Whitaker [221] 提出了一种名为快速迭代方法 (FIM) 的算法来求解并行架构上的 *Eikonal* 方程。

6.2. *Global Optimization*

该子类别包含优化现有初步可行路径的路径规划算法。与第 3.1 节中介绍的 *Local Optimization* 方法不同，*Global Optimization* 方法使生成的路径全局最优，以换取投入更多的计算负载。 Ratliff 等人提出的方法。例如，[222]使用 *Sampling-Based* 方法（例如 RRT 或 PRM）作为第一步。第二步包括使用梯度优化技术从该可行路径逼近最优解。范登伯格等人。 [223] 也从使用 RRT 计算的轨迹开始，随后将 jt 应用于基于微分动态程序的优化过程 -

ming (DP). Plonski et al. [70] used DP to calculate a path in a solar map that dynamically changes, considering that the robot harvests solar energy. Ajanović et al. [224] combined DP with Model Predictive Control (MPC) to calculate energy-minimizing paths. Other techniques include the bang-bang approach [225] and Mixed-Integer Linear Programming (MILP) [226]. Finally, a remarkable approach was proposed by Kogan and Murray [227], who used nonlinear optimization to plan time-optimal paths with a length between 20 and 70 meters.

7. Summary and Conclusions

Table 3 summarizes the main features of each path planning category according to the classification system proposed in this paper (see Figure 1). It analyzes whether the algorithms require a preliminary model of the environment, whether they are deterministic (i.e., they always provide the same solution given the same initial conditions), whether they can tackle dynamic environments and replan, whether they are optimal and if they are complete (i.e., they always return a path if it is feasible). Given the scope of the final path planning application, some algorithms will be more suitable than others. Moreover, the reach of the planner and the replanning capability, i.e., the capability to deal with updates in the environment information, will determine if an algorithm is more suitable for local planning or global planning. Local planning usually requires fast online computation and this reactivity behavior is required to plan new paths in the presence of environmental data changes. Global planning can even be computed offline and aims to generate paths for long traverses, having a static initial environment available.

Table 3. Overview of characteristics of the presented path planning algorithms.

Category	Sub-Category	Preliminary Map Model	Deterministic	Replanning	Optimality	Completeness
Reactive Computing	Reactive Manoeuvre	No [7]	Yes	Yes	Sub-optimal	Not ensured
	Local Optimization	Depends on planner used in first step	Yes	Yes	Sub-optimal	Not ensured
Soft Computing	Evolutionary	Only ACO	No	Yes [3]	Heuristic	Depends (e.g., GA no; PSO, yes) [20]
	Artificial Intelligence	Depends	No	Yes	Heuristic	Not ensured
C-Space Search	Graph Search	Yes [7,11]	Yes [7]	Only incremental	Global restricted to graph	Yes
	Sampling-Based	No	No [7]	Yes	Asymptotical	Probabilistic
Optimal Control	PDE Solving	Yes	Yes	Very rare [212–214]	Globally optimal	Yes
	Global Optimization	Depend on planner used in first step	Yes	Yes [3]	Globally optimal	No [20]

Reactive-Computing-based algorithms seem suitable for local obstacle avoidance path planning as they are easy and cheap to implement. Furthermore, *Reactive Manoeuvre* methods are a good option for scenarios with high uncertainty or when using a robot with very limited sensing options. *Local Optimization* allows one to even consider kinodynamic constraints with TEB, although they do not ensure completeness. Special attention must be given to both subcategories, in order to avoid falling into a local minimum. *Soft Computing* algorithms produce a path using multiple configurable operators, which can be inspired by nature or can be based on fuzzy rules and/or neural networks. They are suitable for problems involving a large number of variables or problems that are difficult to model, such as in highly dynamic environments. With scenarios containing moving elements, in long-range (global path planning) scenarios, the use of *Evolutionary* methods is adequate. The latest *Artificial Intelligence* methods, including the DL and RL methods, still need to be further studied to obtain solid conclusions, as also remarked by Sun et al. [21]. *Artificial Intelligence* methods based on Fuzzy rules or neural networks can be used for fast *Local*

明（DP）。普隆斯基等人。考虑到机器人收集太阳能，[70]使用DP来计算动态变化的太阳图中的路径。阿贾诺维奇ć等人。 [224]将DP与模型预测控制（MPC）相结合来计算能量最小化路径。其他技术包括 bang-bang 方法 [225] 和混合整数线性规划 (MILP) [226]。最后，Kogan 和 Murray [227] 提出了一种非凡的方法，他们使用非线性优化来规划长度在 20 到 70 米之间的时间最优路径。

7. 总结和结论

表3根据本文提出的分类系统总结了每个路径规划类别的主要特征（见图1）。它分析算法是否需要环境的初步模型，它们是否是确定性的（即，它们总是在给定相同初始条件的情况下提供相同的解决方案），它们是否可以处理动态环境并重新规划，它们是否是最优的以及是否完整（即，如果可行，它们总是返回一条路径）。考虑到最终路径规划应用的范围，某些算法将比其他算法更合适。此外，规划器的范围和重新规划能力，即处理环境信息更新的能力，将决定算法是否更适合局部规划或全局规划。局部规划通常需要快速在线计算，并且需要这种反应行为来在环境数据发生变化的情况下规划新路径。全局规划甚至可以离线计算，旨在生成长距离遍历的路径，并具有可用的静态初始环境。

表 3. 所提出的路径规划算法的特征概述。

Category	Sub-Category	Preliminary Map Model	Deterministic	Replanning	Optimality	Completeness
Reactive Computing	Reactive Manoeuvre	No [7]	Yes	Yes	Sub-optimal	Not ensured
	Local Optimization	Depends on planner used in first step	Yes	Yes	Sub-optimal	Not ensured
Soft Computing	Evolutionary	Only ACO	No	Yes [3]	Heuristic	Depends (e.g., GA no; PSO, yes) [20]
	Artificial Intelligence	Depends	No	Yes	Heuristic	Not ensured
C-Space Search	Graph Search	Yes [7,11]	Yes [7]	Only incremental	Global restricted to graph	Yes
	Sampling-Based	No	No [7]	Yes	Asymptotical	Probabilistic
Optimal Control	PDE Solving	Yes	Yes	Very rare [212–214]	Globally optimal	Yes
	Global Optimization	Depend on planner used in first step	Yes	Yes [3]	Globally optimal	No [20]

*Reactive-Computing*基于 的算法似乎适合局部避障路径规划，因为它们易于实现且成本低廉。此外，对于高度不确定性的场景或使用传感选项非常有限的机器人时，*Reactive Manoeuvre* 方法是一个不错的选择。 *Local Optimization* 甚至允许人们考虑 TEB 的运动动力学约束，尽管它们不能确保完整性。必须特别注意这两个子类别，以避免陷入局部最小值。 *Soft Computing* 算法使用多个可配置运算符生成一条路径，这些运算符可以受到自然的启发，也可以基于模糊规则和/或神经网络。它们适用于涉及大量变量的问题或难以建模的问题，例如在高度动态的环境中。对于包含移动元素的场景，在远距离（全局路径规划）场景中，使用*Evolutionary*方法就足够了。 Sun 等人也指出，最新的 *Artificial Intelligence* 方法，包括 DL 和 RL 方法，仍需要进一步研究才能获得可靠的结论。[21]。基于模糊规则或神经网络的 *Artificial Intelligence* 方法可用于快速 *Local*

Planning as an alternative to *Reactive Manoeuvre* methods. *C-Space Search* algorithms make use of samples to represent the different configurations of the robot. These samples can be provided beforehand in the form of a graph or they can be dynamically created. *Graph Search* algorithms are suitable for global path planning considering advanced graphs such as visibility graphs or space-lattice graphs, at the expense of investing time into building them (something that is admissible for offline planning). Nevertheless, it scales poorly with problems of high dimensions, which justifies the use of *Sampling-based* algorithms instead. *Sampling-based* algorithms have also proved useful for this kind of manoeuvre and problems with a high number of dimensions. *Optimal Control* algorithms are outstanding for obtaining globally optimal results. *PDE-Solving*-based algorithms depend heavily on the formulated PDE, which can be based on isotropic or anisotropic cost functions and can work with a map model in the form of a grid. *Global Optimization* algorithms have to start with an already-defined path an adapt it according to the robot's locomotion restrictions. *PDE Solving* algorithms are adequate for the off-line computation of long traverses given low-uncertain static scenarios, as they provide optimal paths without relying on replanning. Finally, it is important to note that all these planners rely on the available information describing the environment and the robot. This information must be modeled as accurately as possible to improve the results of the path planner.

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Abbreviations

The following abbreviations are used in this manuscript:

ACO	Ant Colony Optimizer
AIT*	Adaptively Informed Tree
APF	Artificial Potential Fields
ABIT*	Advanced Batch Informed Trees
BIT*	Batch Informed Trees
D*	Dynamic A*
DEM	Digital Elevation Map
DL	Deep Learning
DP	Dynamic Programming
DPP	Dynamic Programming Principle
DWA	Dynamic Window Approach
FIM	Fast Iterative Method
FMM	Fast Marching Method
FMT*	Fast Marching Tree
FMS	Fast Marching Square

Planning 作为 *Reactive Manoeuvre* 方法的替代方法。 *C-Space Search* 算法利用样本来表示机器人的不同配置。这些样本可以预先以图表的形式提供，也可以动态创建。 *Graph Search* 算法适用于考虑高级图（例如可见性图或空间点阵图）的全局路径规划，但代价是花费时间来构建它们（离线规划是允许的）。然而，它对于高维度问题的扩展性很差，这证明了使用 *Sampling-based* 算法是合理的。 *Sampling-based* 算法也被证明对于此类操作和高维问题很有用。 *Optimal Control* 算法在获得全局最优结果方面表现出色。基于 *PDE-Solving* 的算法在很大程度上依赖于公式化的偏微分方程，它可以基于各向同性或各向异性成本函数，并且可以与网格形式的地图模型一起使用。 *Global Optimization* 算法必须从已经定义的路径开始，并根据机器人的运动限制对其进行调整。 *PDE Solving* 算法足以在给定低不确定性静态场景的情况下进行长遍历的离线计算，因为它们无需依赖重新规划即可提供最佳路径。最后，值得注意的是，所有这些规划者都依赖于描述环境和机器人的可用信息。必须尽可能准确地对该信息进行建模，以改进路径规划器的结果。

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缩写

本手稿中使用了以下缩写：

ACO 蚁群优化器 AIT* 自适应通知树 APF 人工势场 ABIT* 高级批量通知树 BIT* 批量通知树 D* 动态 A* DEM 数字高程图 DL 深度学习 DP 动态规划 DPP 动态规划原理 DWA 动态窗口法 FIM 快速迭代法 FMM 快速行进法 FMT* 快速行进树 FMS 快速行进方

FSM	Fast Sweeping Method
FM*	Heuristic Fast Marching
GMT*	Group Marching Tree
HJB	Hamilton-Jacobi-Bellman
LPA*	Lifelong Planning A*
MPC	Model Predictive Control
OUM	Ordered Upwind Method
PDE	Partial Derivative Equation
PRM	Probabilistic Roadmap Method
PSO	Particle Swarm Optimizer
RABIT*	Regionally Accelerated Batch Informed Trees
RDT	Rapidly Deterministic Tree
RL	Reinforcement Learning
RRT	Rapidly Random Tree
RRT*	Heuristic Rapidly Random Tree
TEB	Timed Elastic Bands
VFH	Vector Field Histogram

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FSM 快速扫描方法 FM* 启发式快速行进 GMT* 群体行进树 HJB Hamilton-Jacobi-Bellman LPA* 终身规划 A* MPC 模型预测控制 OUM 有序迎风法 PDE 偏导方程 PRM 概率路线图方法 PSO 粒子群优化器 RABIT* 区域加速批量知情树 RDT 快速确定性树 RL 强化学习 RRT快速随机树 RRT* 启发式快速随机树 TEB 定时弹性带 VFH 矢量场直方图

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